

计算机视觉

第9章 图像生成

软件学院：刘春

Ref: C280, Computer Vision
Prof. Trevor Darrell

本章内容

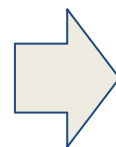
- 9.1.生成对抗网络定义
- 9.2.鉴别模型与生成模型
- 9.3.生成对抗式网络
- 9.4.条件GAN
- 9.5 *. cycleGAN与styleGAN

目前我们已经学习了？



目前我们已经学习了？

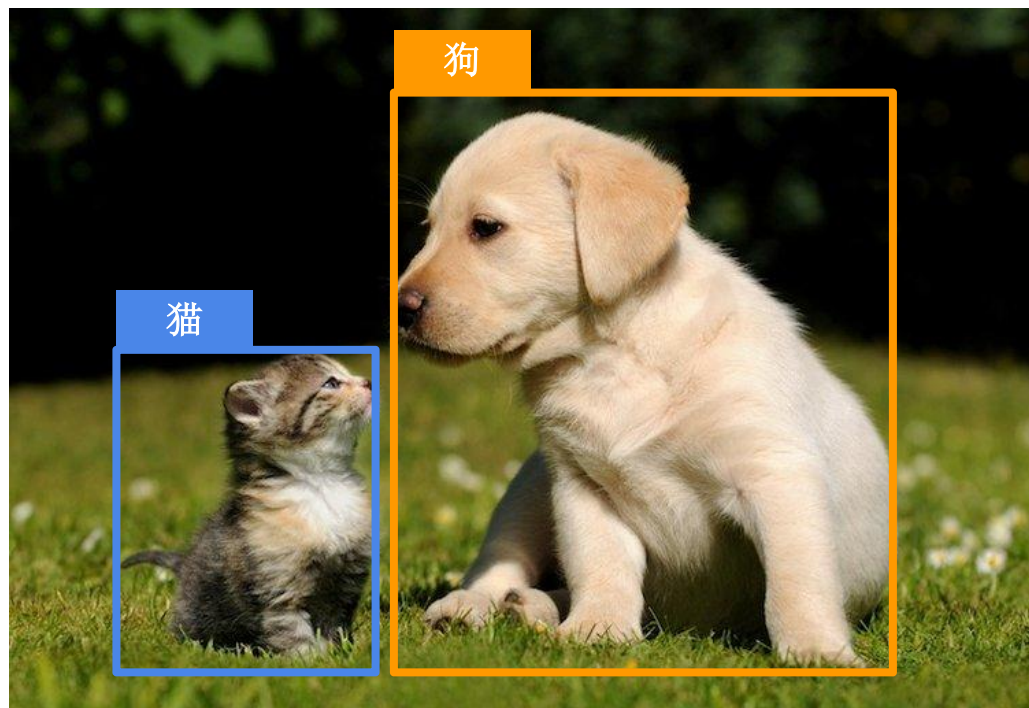
分类



猫
狗
室外
...

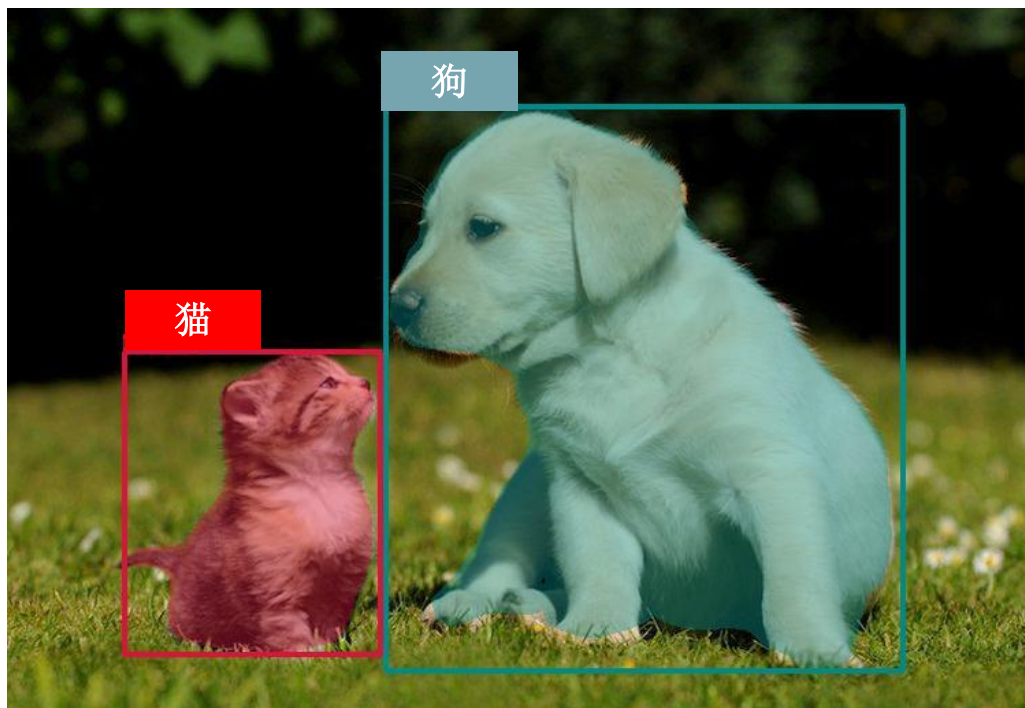
目前我们已经学习了？

检测



目前我们已经学习了？

分割

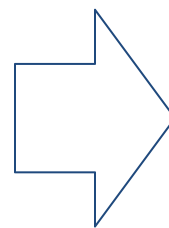


- 区域分割
- 实例分割
- 语义分割
- 全景分割

目前我们已经学习了?

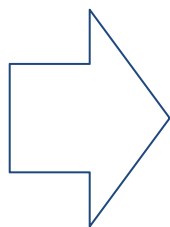


X
数据



Y
标签

今天我们的关注点

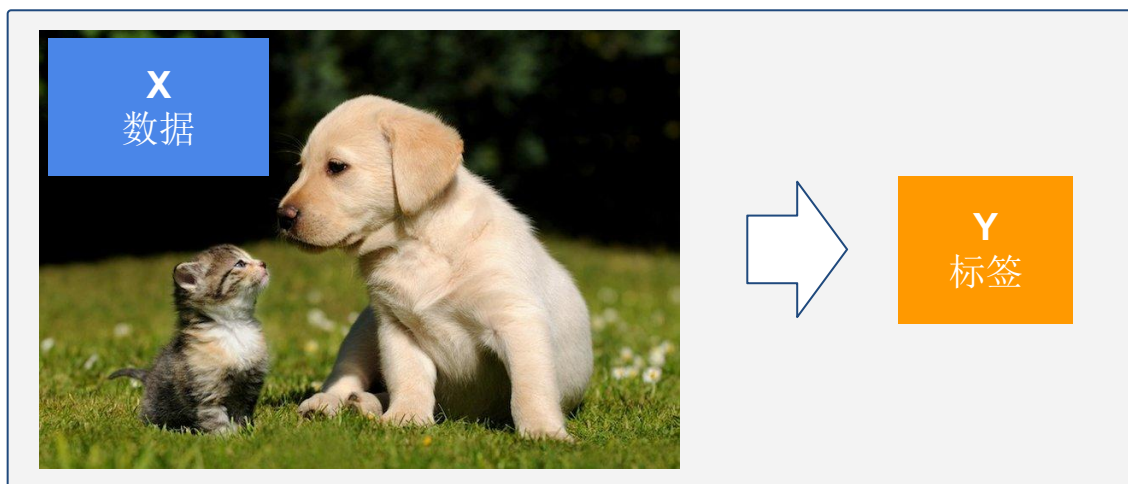


今天我们的关注点

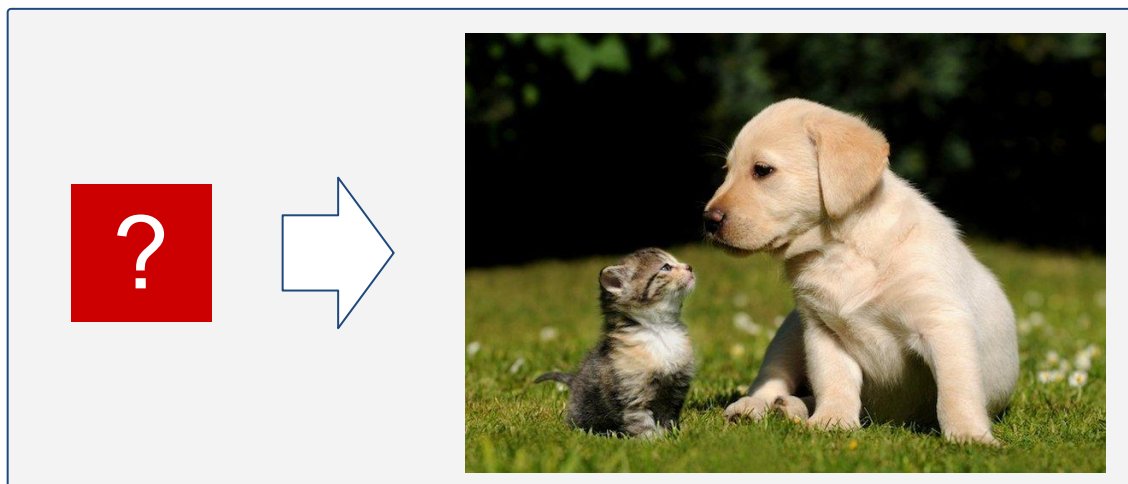


GANPaint: <http://gandissect.res.ibm.com/ganpaint.html?project=churchoutdoor&layer=layer4>

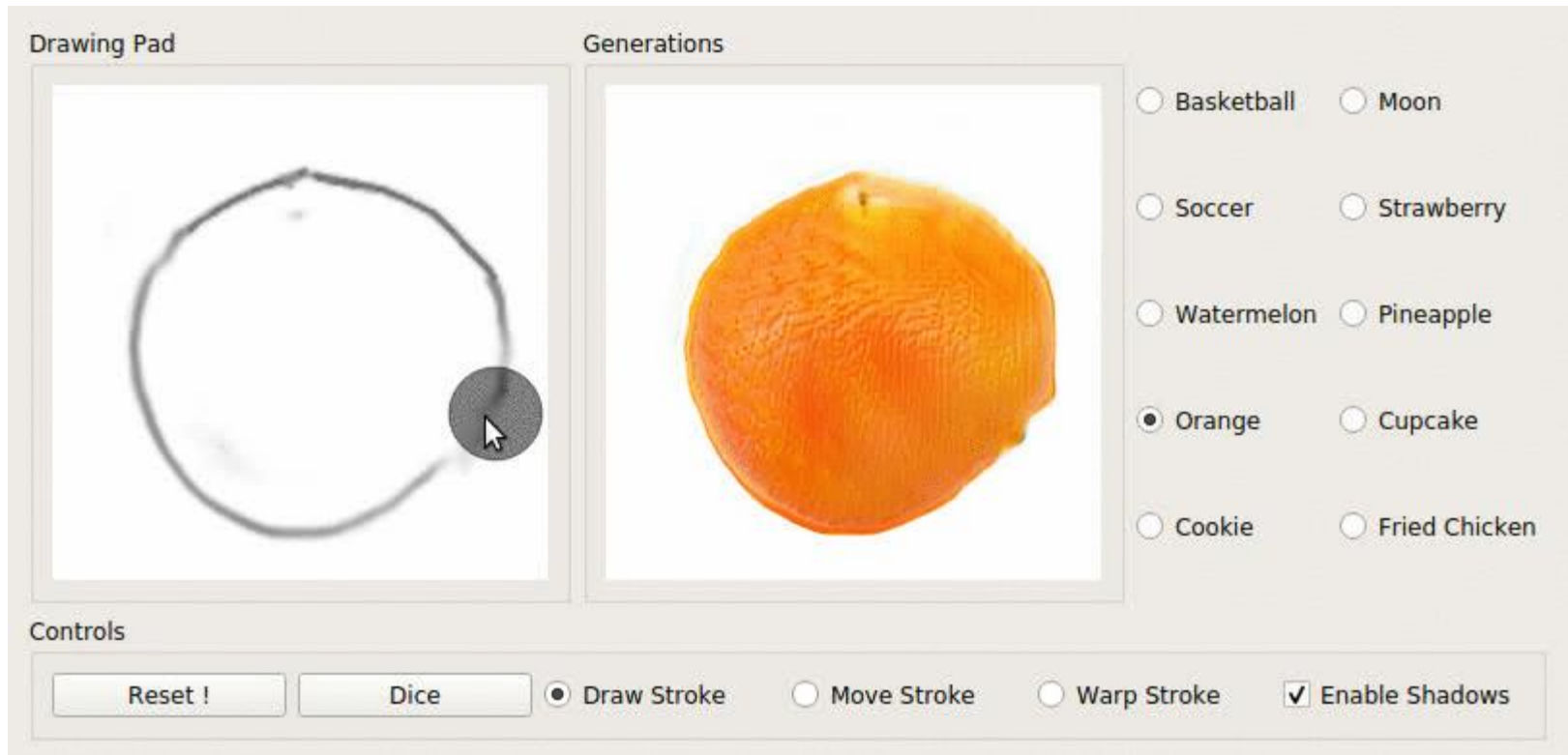
鉴别



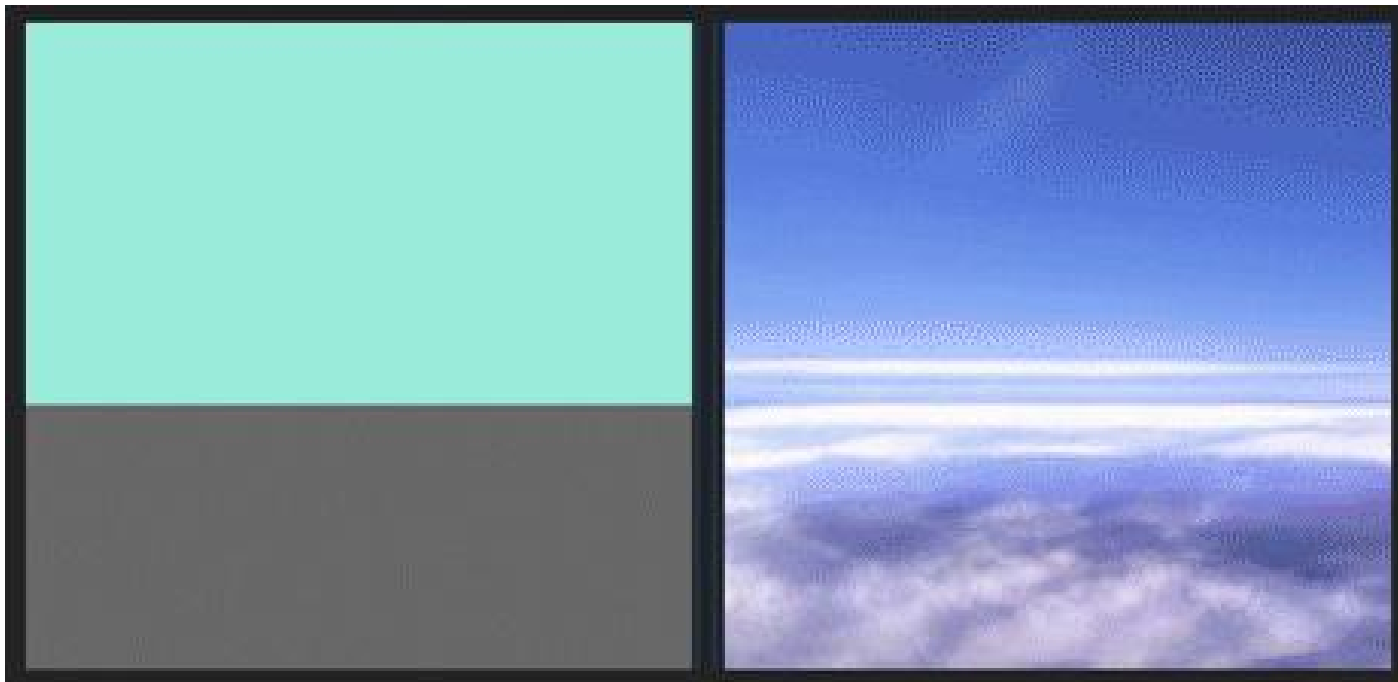
生成



图像仿真



图像仿真



图像PS



<https://www.nvidia.com/en-us/research/ai-playground/>

自动驾驶场景生成

Input labels

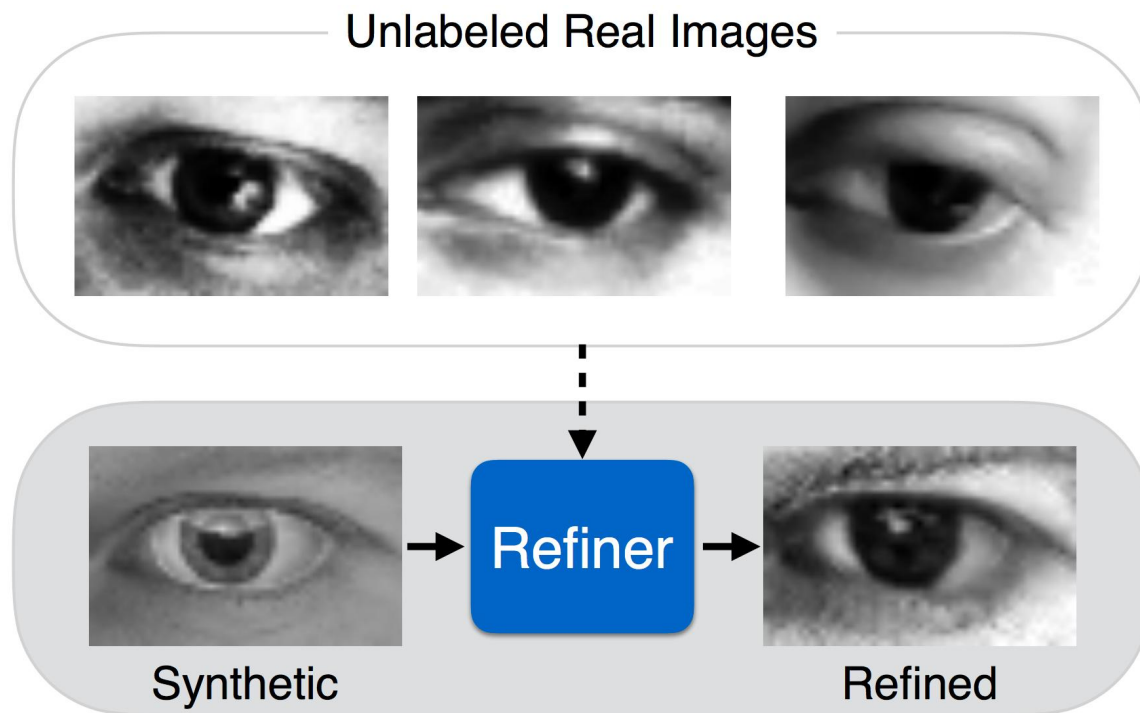


Synthesized image



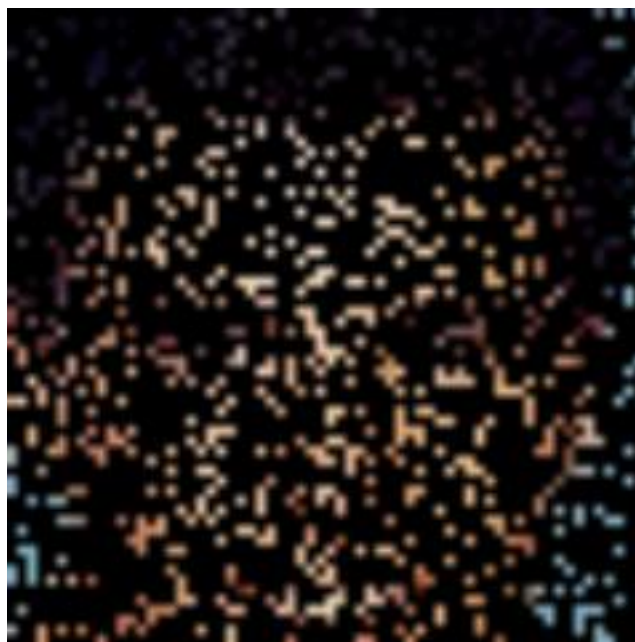
Wang, Ting-Chun, et al. "[High-resolution image synthesis and semantic manipulation with conditional gans](#)",
CVPR 2018.

GANs 训练数据生成



(Shrivastava et al.,
2016)

图像复原



(Yeh et al.,
2016)

图像复原



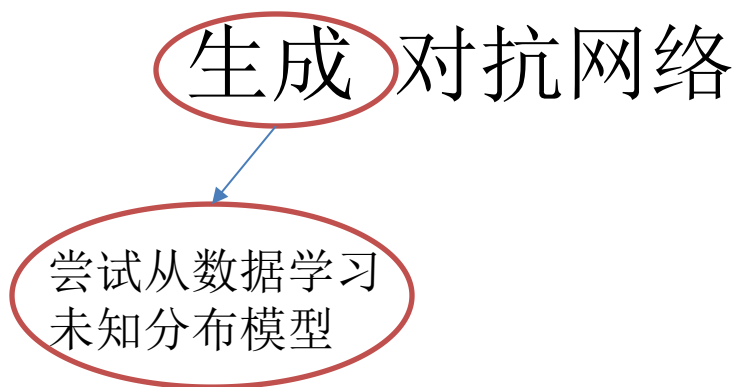
(Yeh et al.,
2016)

图形到图像



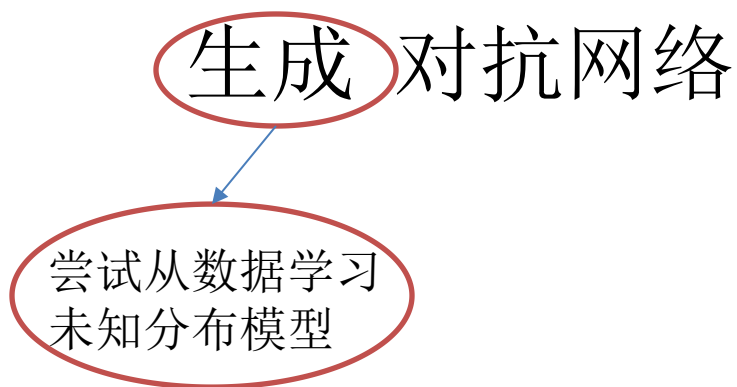
生成对抗网络 (GAN)

- 什么是生成对抗网络？



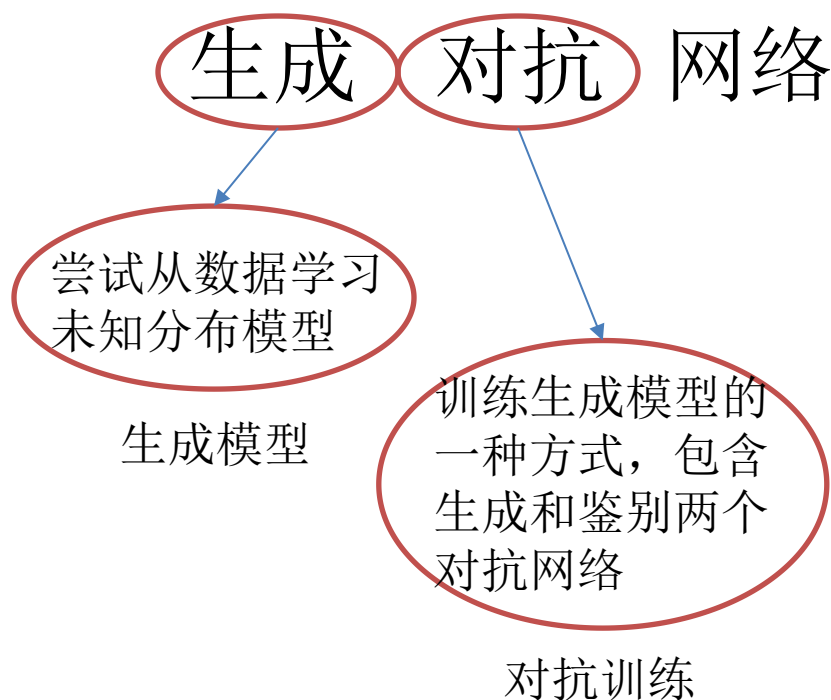
生成对抗网络 (GAN)

- 什么是生成对抗网络？



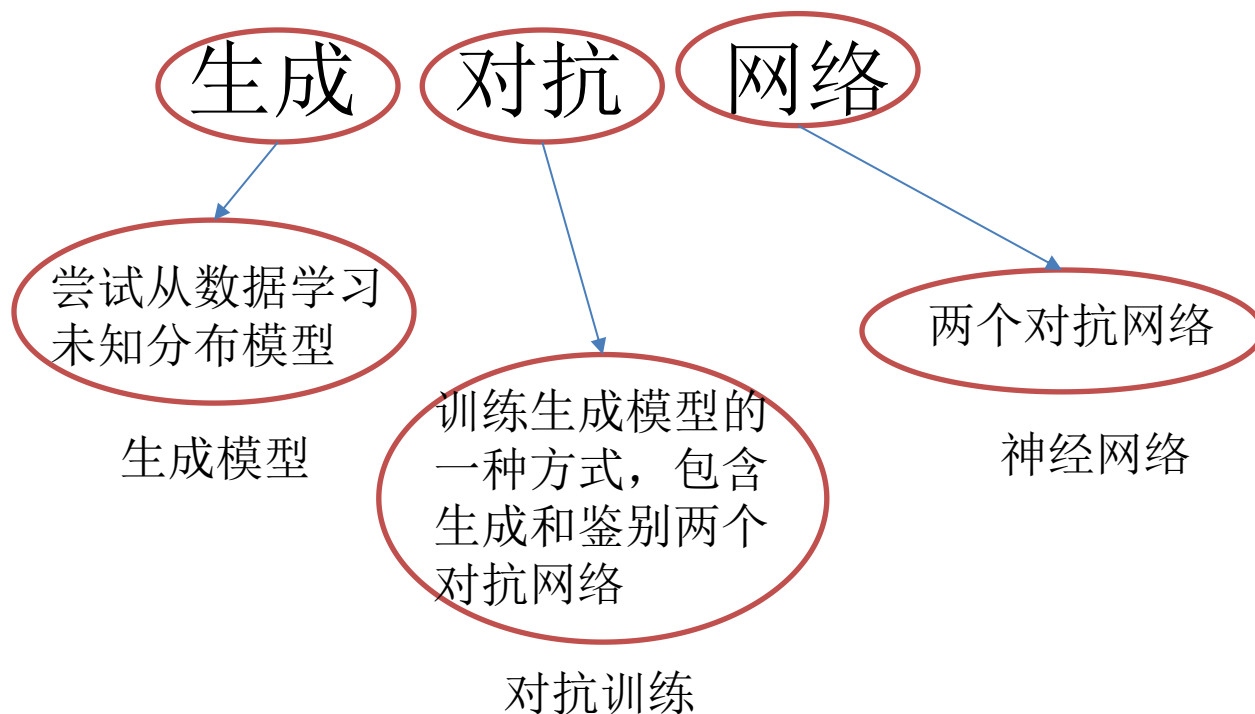
生成对抗网络（GAN）

- 什么是生成对抗网络？



生成对抗网络（GAN）

- 什么是生成对抗网络？



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生成模型与鉴别模型

- 生成模型

学习概率分布 $P(x)$ 或 $P(x, y)$

- 鉴别模型

学习概率分布 $P(y|x)$

数据 x

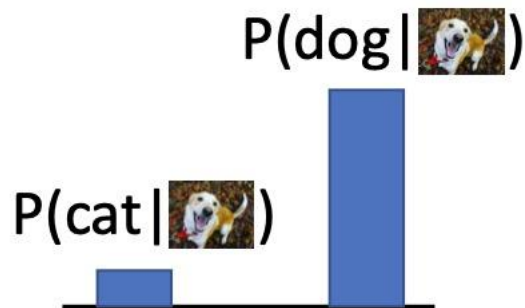
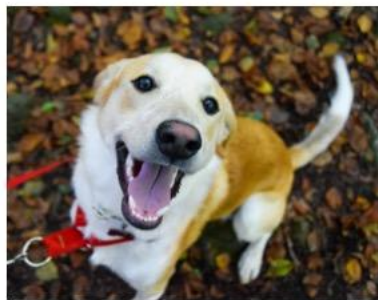
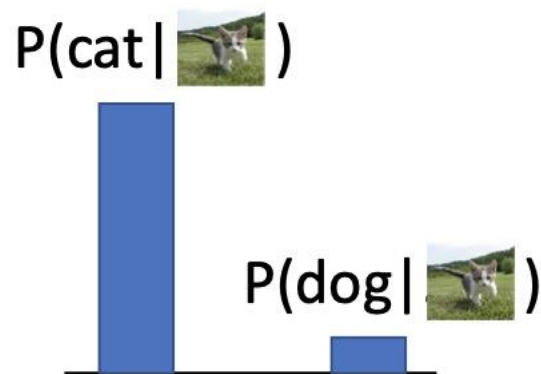


类别 y

猫

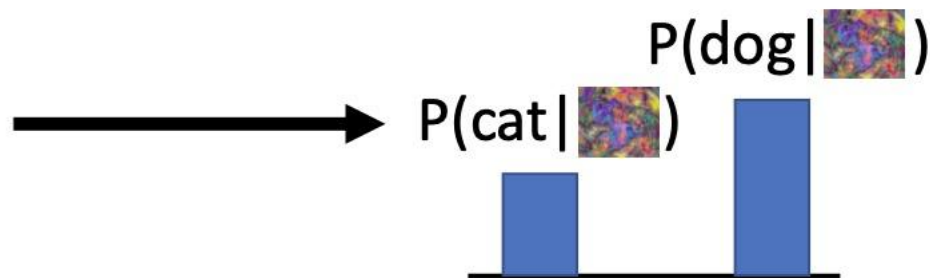
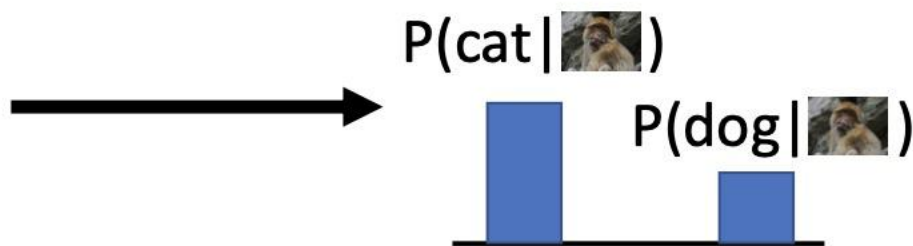
生成模型与鉴别模型

- 鉴别模型—图片之间没有比较



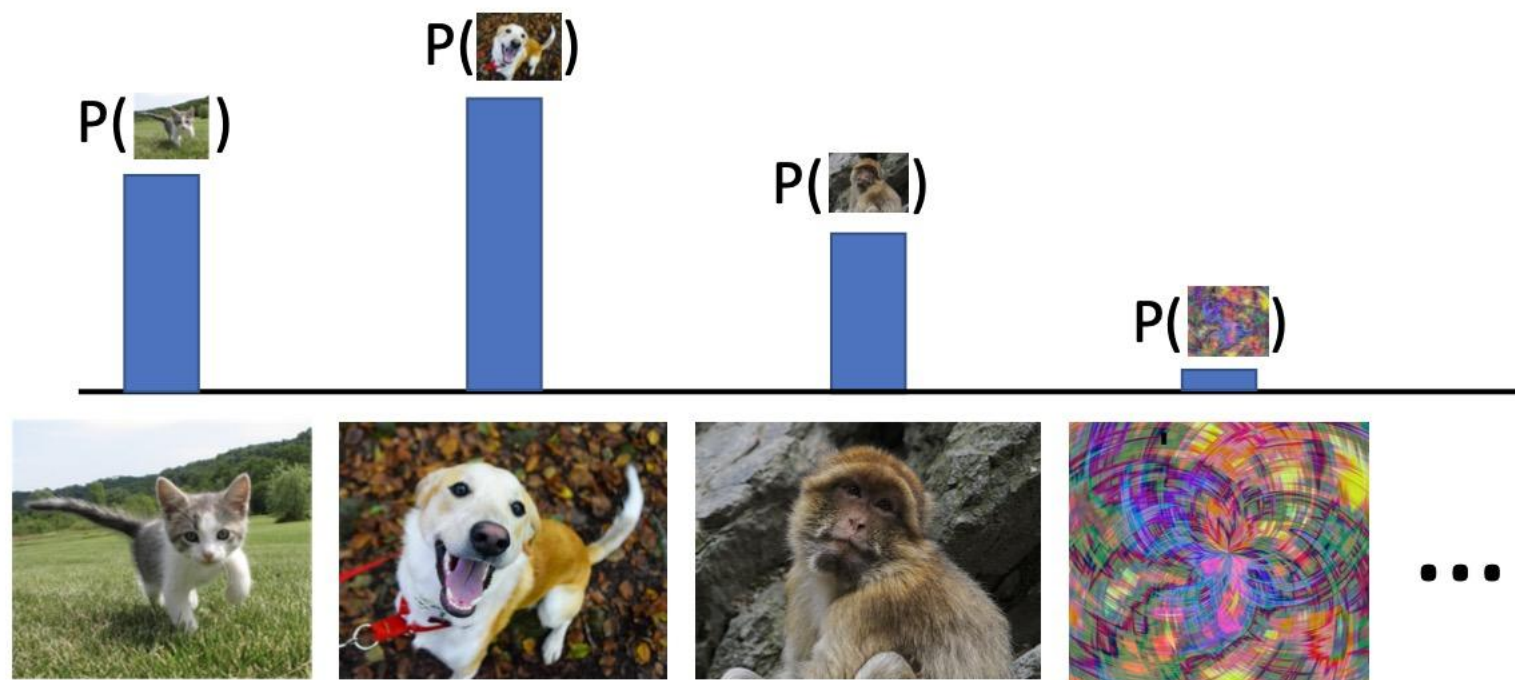
生成模型与鉴别模型

- 鉴别模型—无法处理异常输入



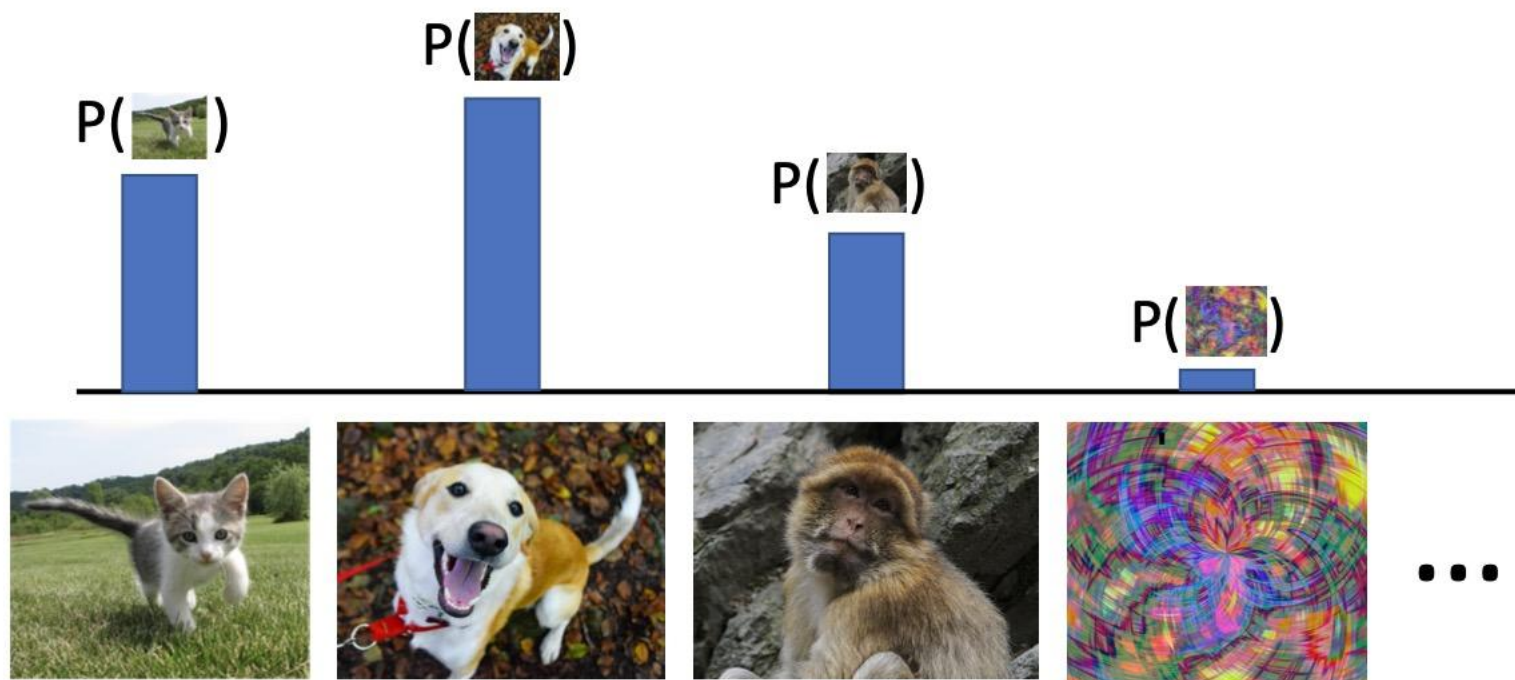
生成模型与鉴别模型

- 生成模型—每张图片计算概率，相互比较



生成模型与鉴别模型

- 生成模型—需要深度理解图片，狗站立还是坐下等图像所有的属性



- 能处理异常输入，概率小剔除

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如何定义生成图片的损失？

输入



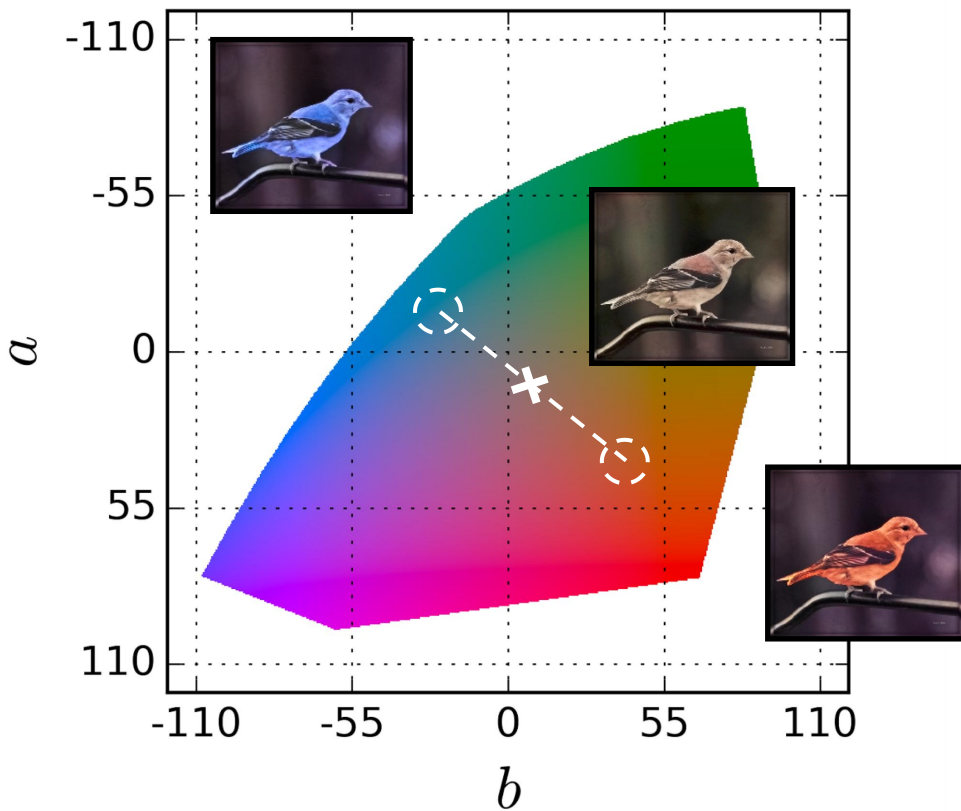
输出



真实分布



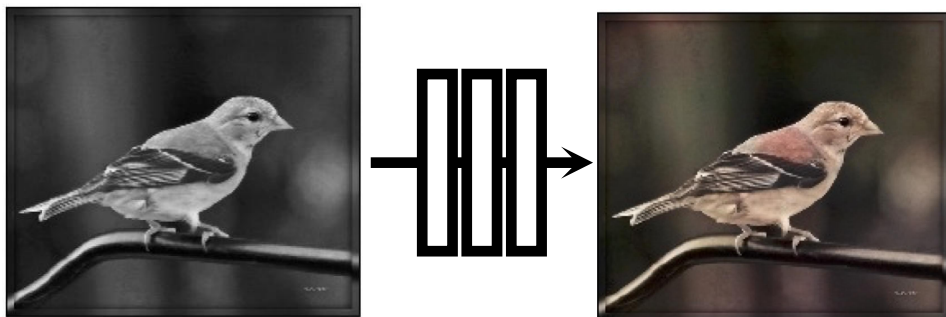
L2 损失？



L2 距离?

如何定义损失?

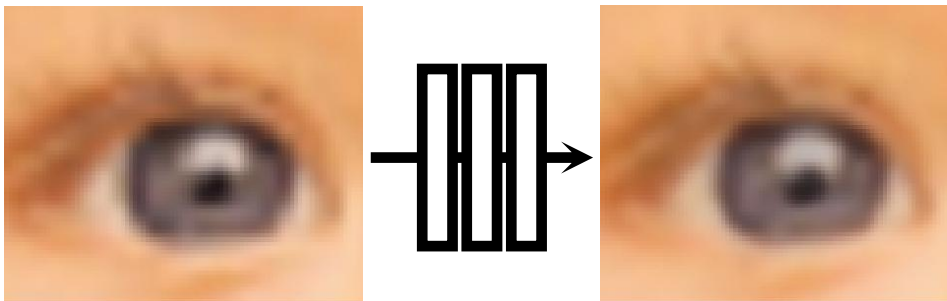
图像着色



L2 损失

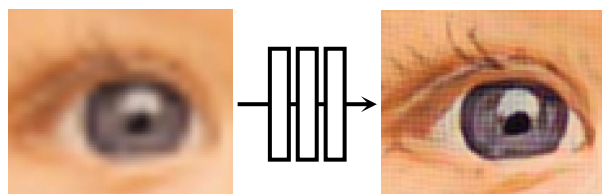
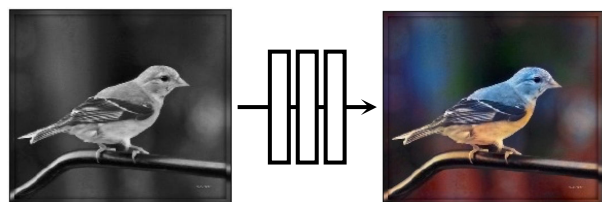
[Colorful Image Colorization](#), Zhang, Isola, Efros, ECCV 2016

超分辨



L2 损失

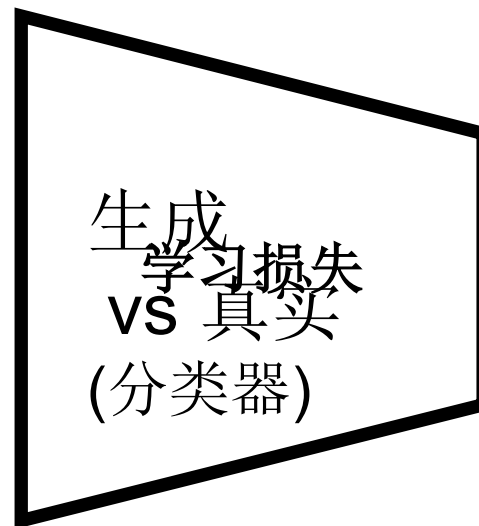
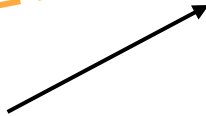
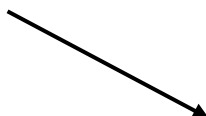
[Perceptual Losses for Real-Time Style Transfer and Super-Resolution](#),
Johnson, Alahi, Li, ECCV 2016



⋮

⋮

生成数据



真
或
假

真实数据



[Goodfellow, Pouget-Abadie, Mirza, Xu, Warde-Farley, Ozair, Courville, Bengio 2014]

[Goodfellow, Pouget-Abadie, Mirza, Xu, Warde-Farley, Ozair, Courville, Bengio 2014]

制假钞

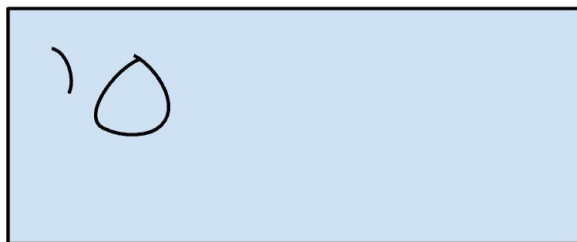


假钞生成器

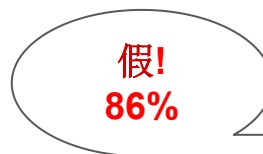
第一次尝试



假钞生成器



假



鉴别



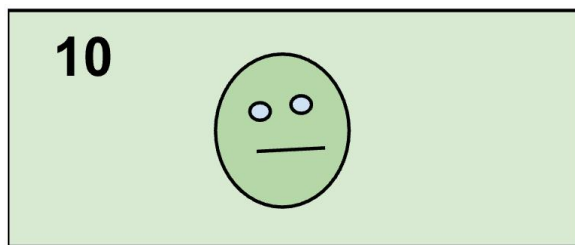
真



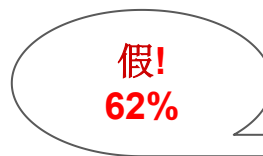
一些尝试之后..



假钞生成器



假



鉴别



真



许多尝试之后..



假钞生成器



假

真!
87%



鉴别

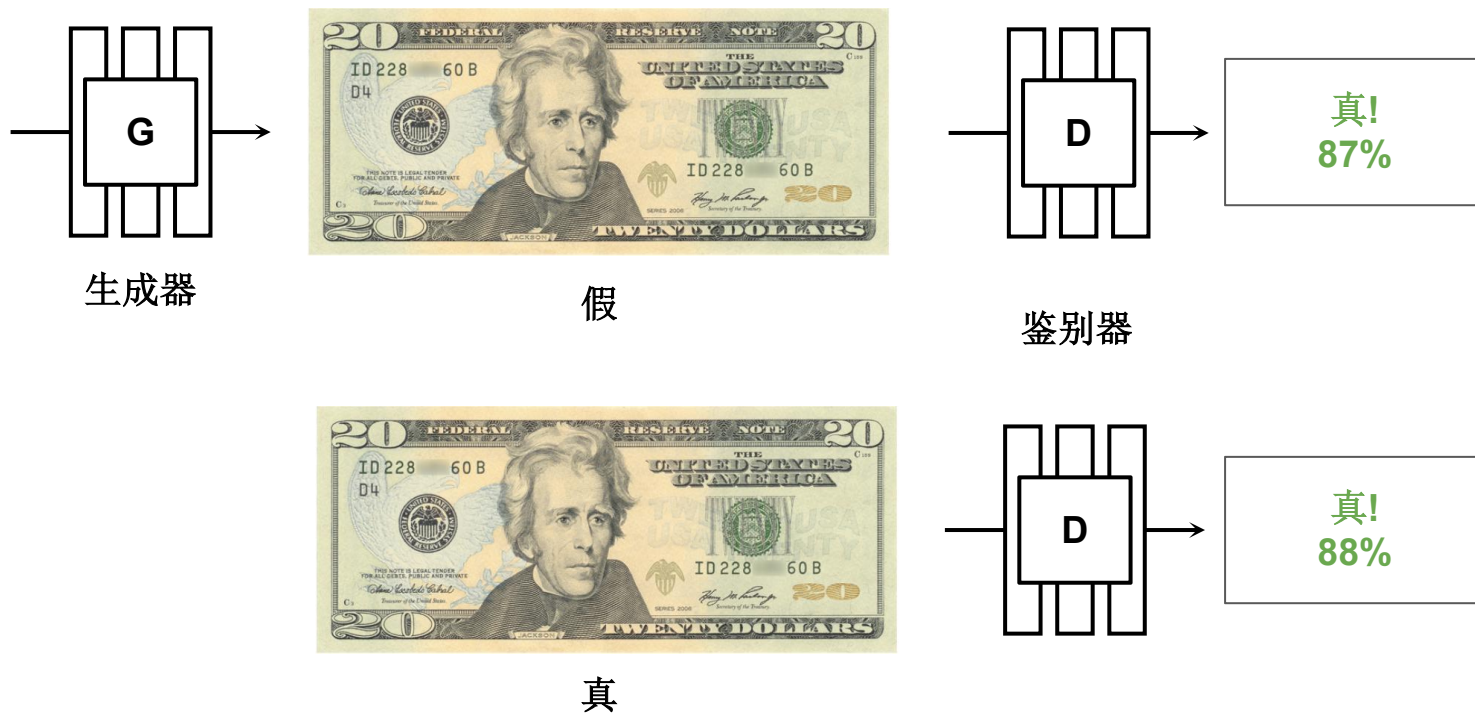


真

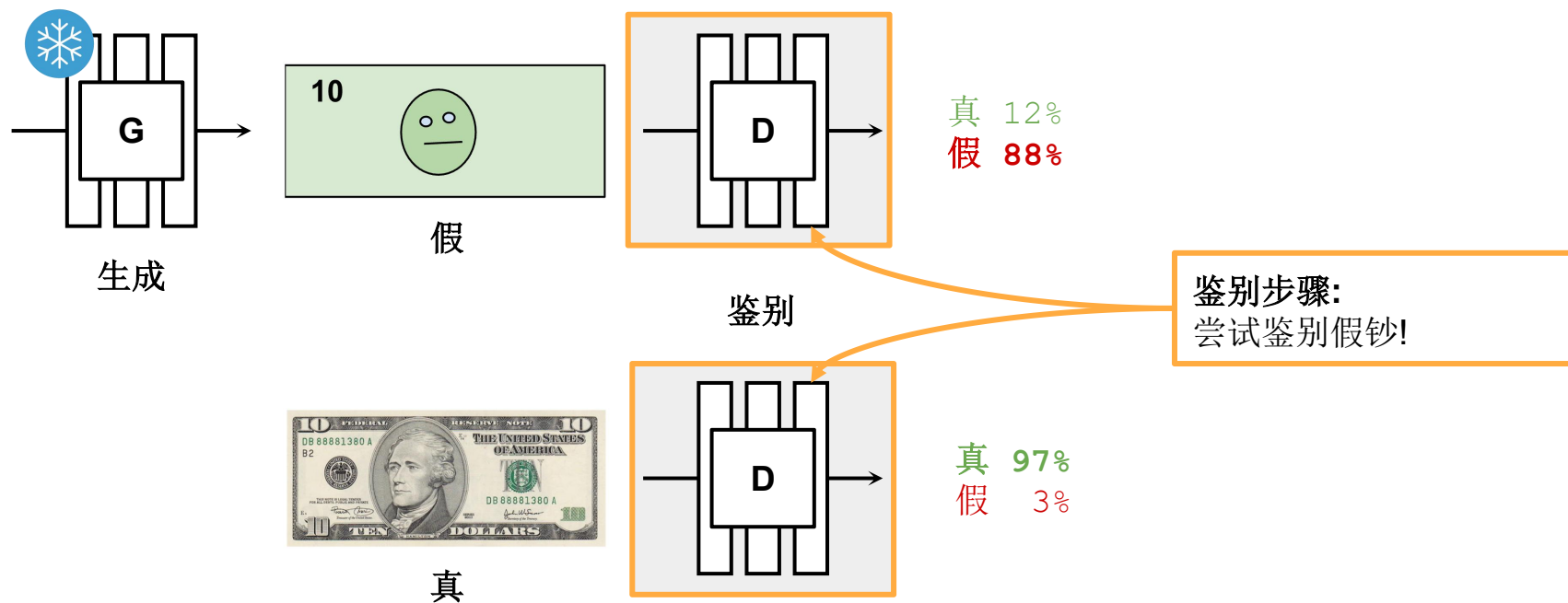
真!
88%

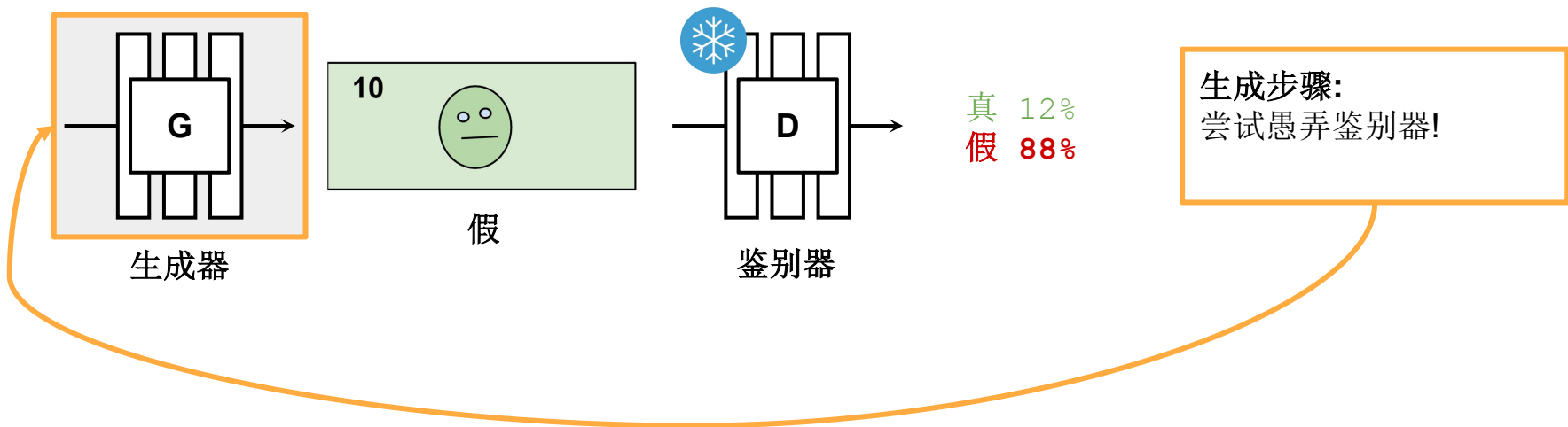


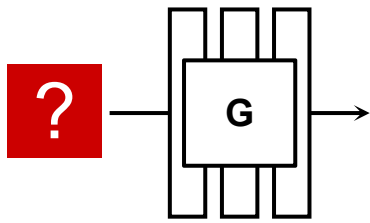
生成对抗网络(GAN)



生成对抗网络(GAN)

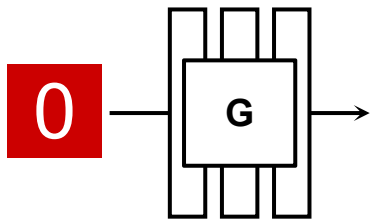






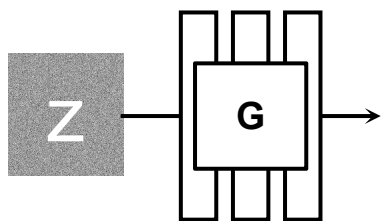
生成器





生成





生成



生成对抗网络

- 生成模型:

由已知数据 x_i 获得数据分布 $p_{data}(x)$

- 想法: 引入隐藏变量 z , 先验分布为 $p(z)$

按 $p(z)$ 采样 z 通过一个生成网络 $x = G(z)$

由此输出 x 是一个分布为 p_G 的样本, 希望 $p_G = p_{data}$

生成对抗网络

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由已知数据 x_i 获得数据分布 $p_{data}(x)$

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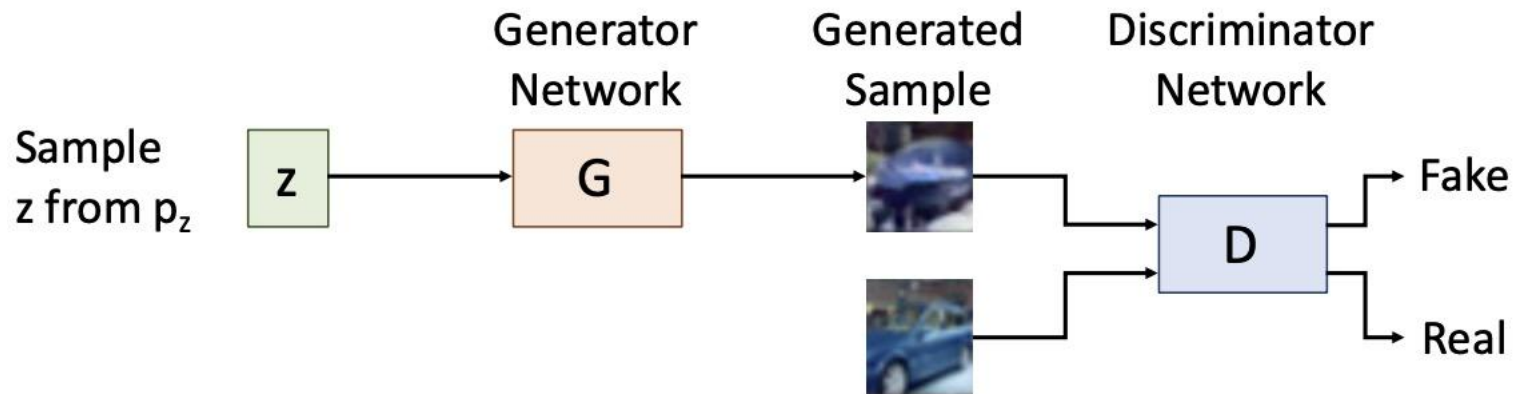
由此输出 x 是一个分布为 p_G 的样本, 希望 $p_G = p_{data}$



训练一个网络 G 将 z 转换为分布为 p_G 的 x

生成对抗网络

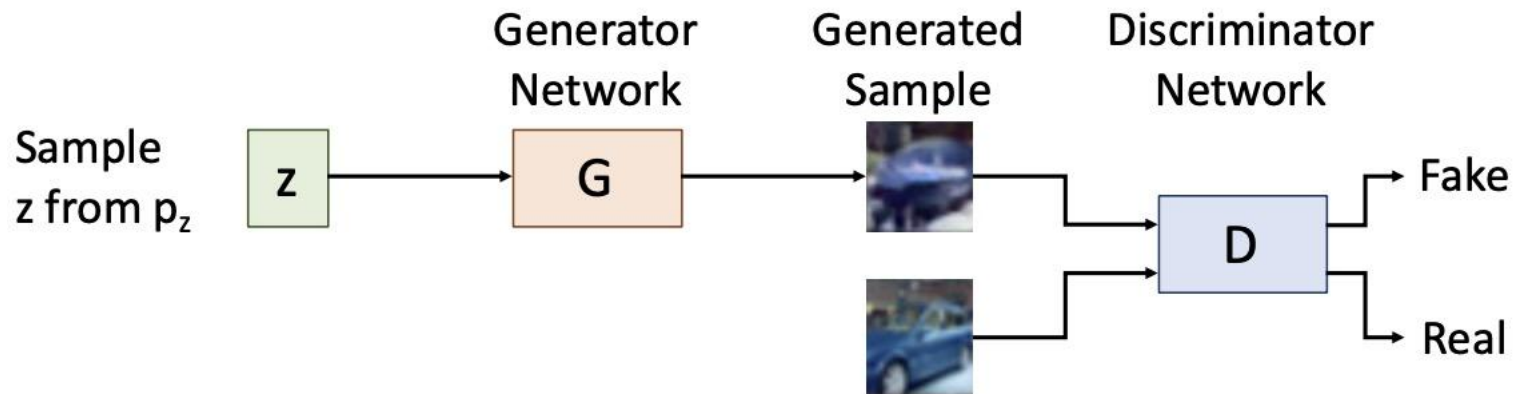
- 如何定义损失



- 使用鉴别网络计算损失
训练鉴别器D用于鉴别生成数据为真或假

生成对抗网络

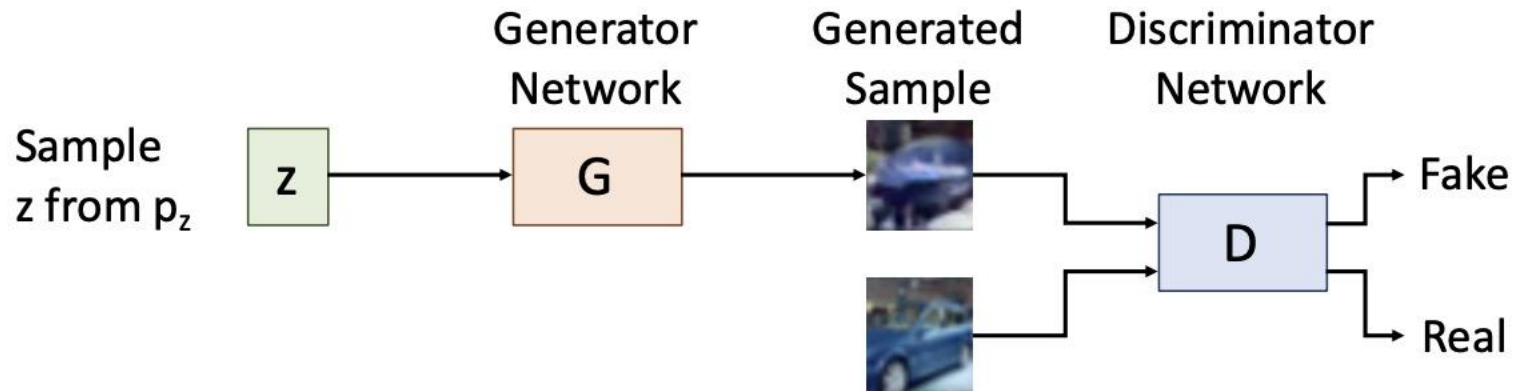
- 同时训练生成网络和鉴别网络



生成对抗网络

- 定义一个纳什均衡目标函数进行联合训练

$$\min_G \max_D \left(E_{x \sim p_{data}} [\log D(x)] + E_{z \sim p(z)} [\log (1 - D(G(z)))] \right)$$



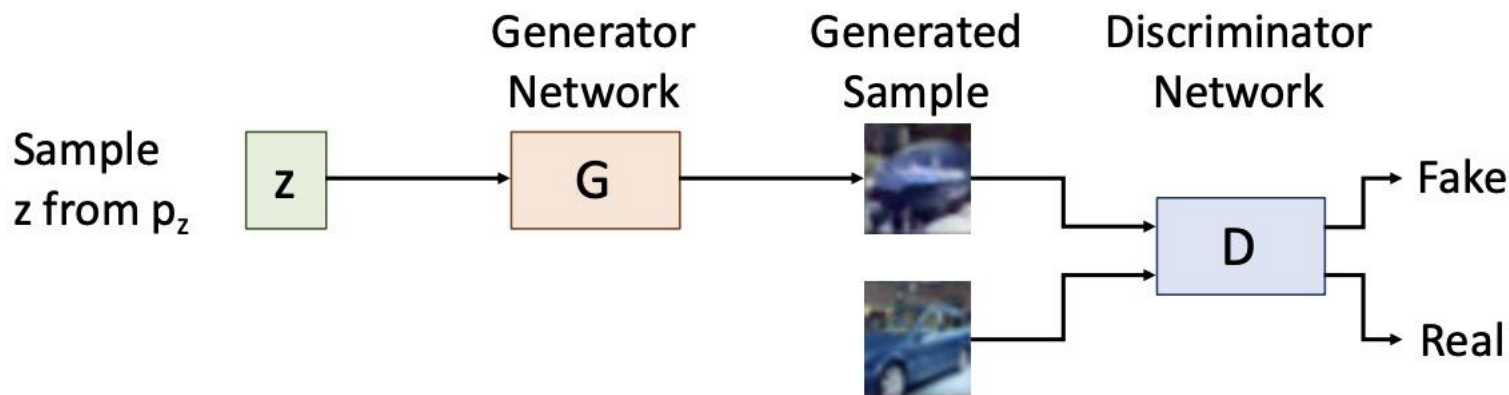
生成对抗网络

- 定义一个纳什均衡目标函数进行联合训练

鉴别器希望对于
真实数据 $D(x)=1$

鉴别器希望对于
生成数据 $D(x)=0$

$$\min_G \max_D \left(E_{x \sim p_{data}} [\log D(x)] + E_{z \sim p(z)} [\log (1 - D(G(z)))] \right)$$



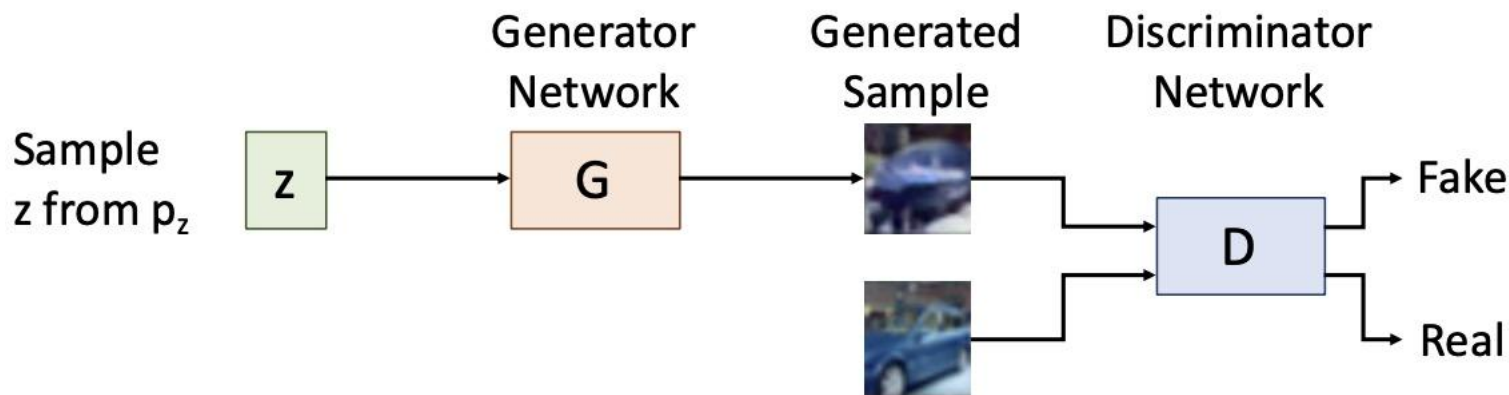
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生成对抗网络

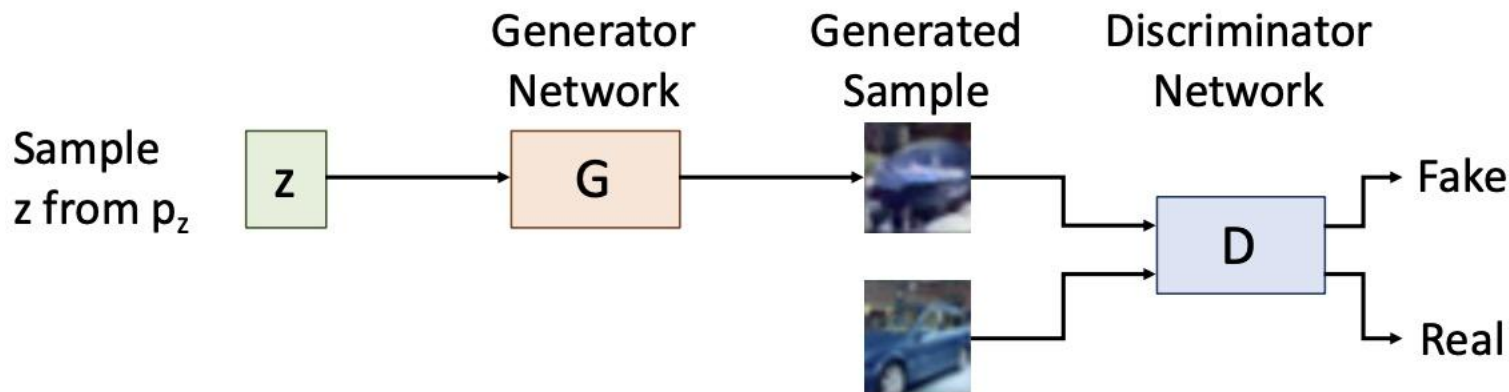
- 定义一个纳什均衡目标函数进行联合训练

鉴别器希望对于
真实数据 $D(x)=1$

鉴别器希望对于
生成数据 $D(x)=0$

$$\min_G \max_D \left(E_{x \sim p_{data}} [\log D(x)] + E_{z \sim p(z)} \left[\log (1 - D(G(z))) \right] \right)$$

生成器希望对于
生成数据 $D(x)=1$



生成对抗网络

- 定义一个纳什均衡目标函数进行联合训练
训练G和D通过交替梯度更新迭代

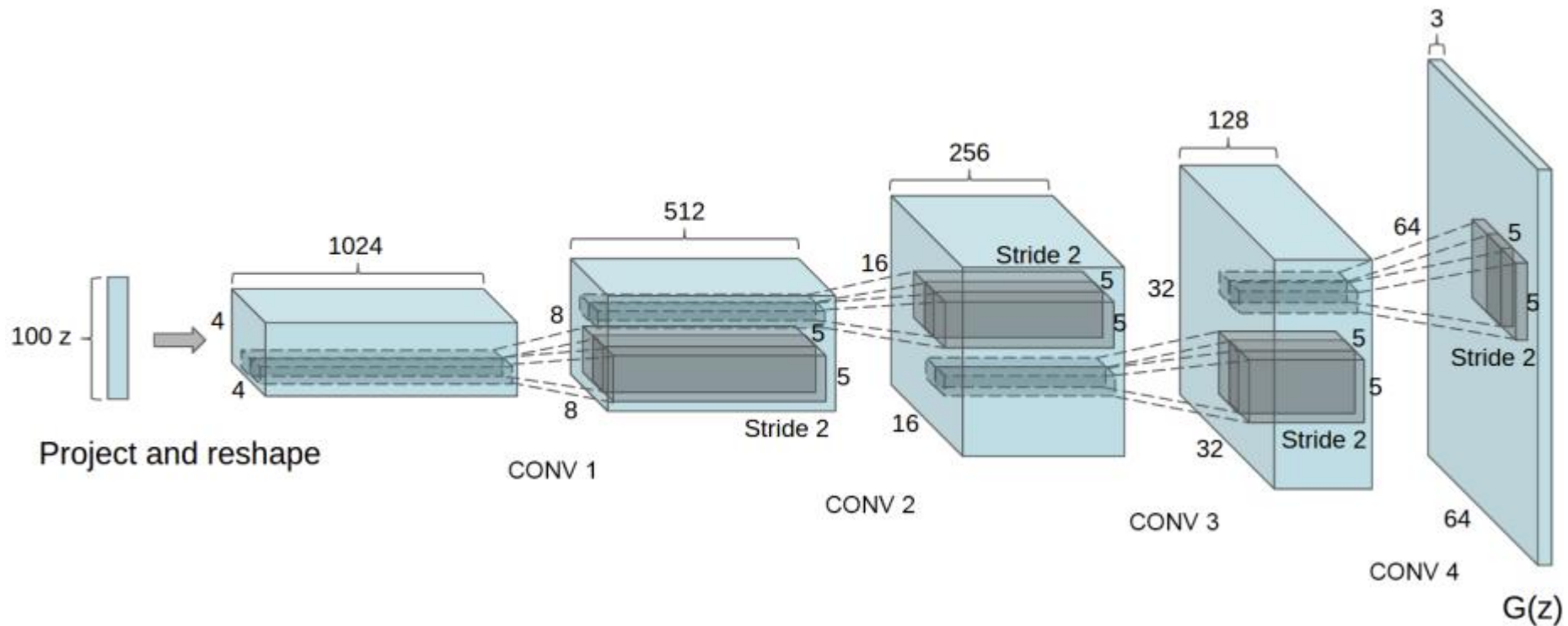
$$\min_{\mathbf{G}} \max_{\mathbf{D}} \left(E_{x \sim p_{data}} [\log \mathbf{D}(x)] + E_{z \sim p(z)} [\log (1 - \mathbf{D}(\mathbf{G}(z)))] \right)$$

$$= \min_{\mathbf{G}} \max_{\mathbf{D}} V(\mathbf{G}, \mathbf{D})$$

For t in 1, ... T:

1. (Update $\mathbf{D}$) $\mathbf{D} = \mathbf{D} + \alpha_{\mathbf{D}} \frac{\partial V}{\partial \mathbf{D}}$
2. (Update $\mathbf{G}$) $\mathbf{G} = \mathbf{G} - \alpha_{\mathbf{G}} \frac{\partial V}{\partial \mathbf{G}}$

DCGAN: Deep Convolutional Generative Adversarial Networks

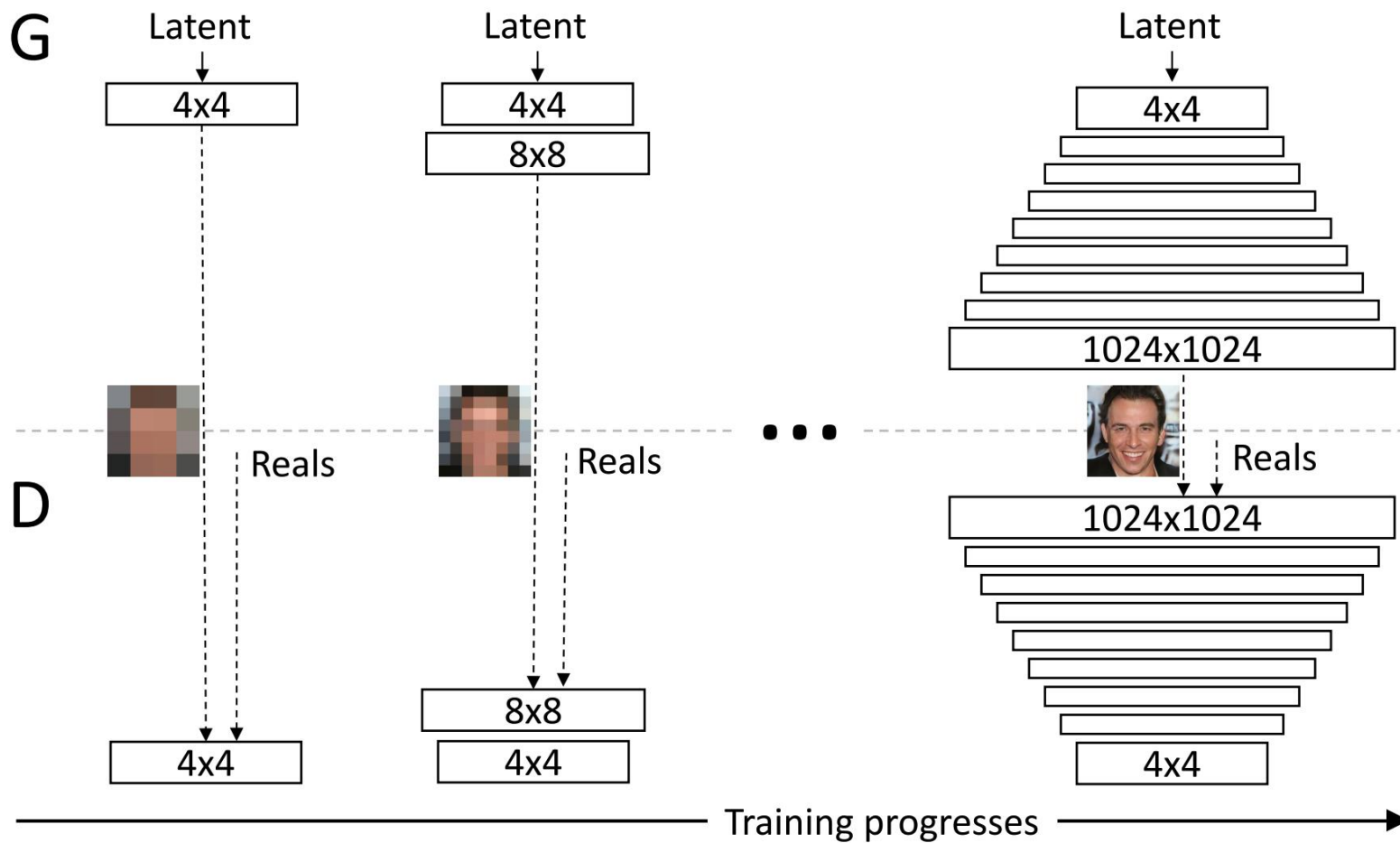


[Radford et. al., Unsupervised Representation Learning with Deep Convolutional Generative Adversarial Networks]⁵³

Training Images





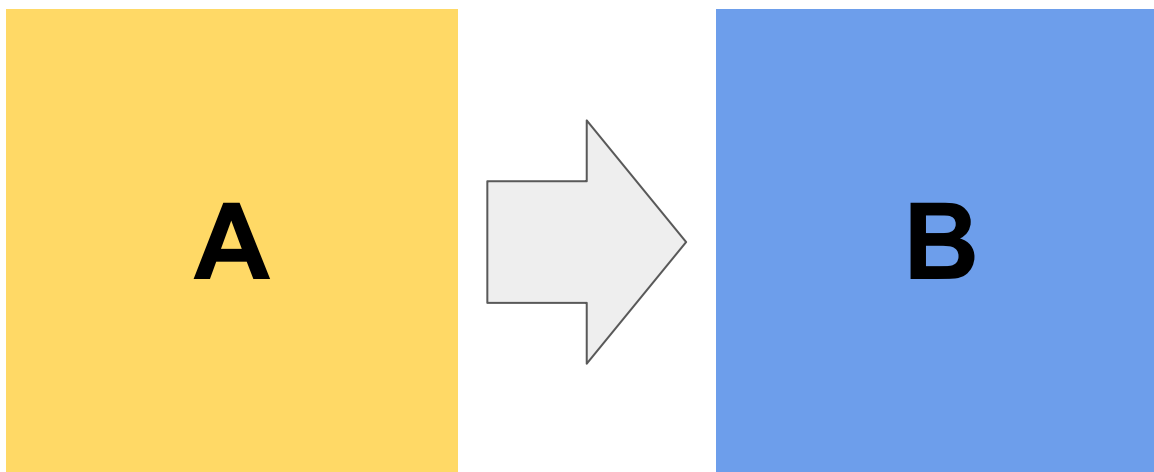


[Karras et. al., Progressive Growing of GANs for Improved Quality, Stability, and Variation, ICLR 2018]

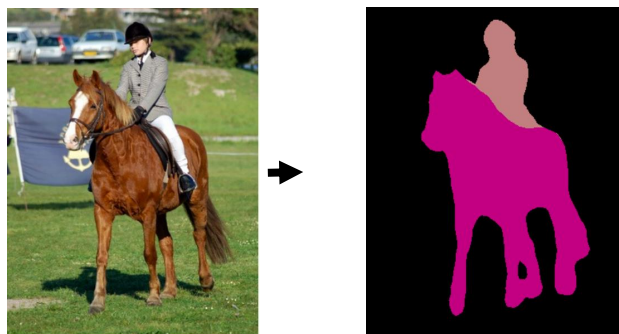
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图像转换

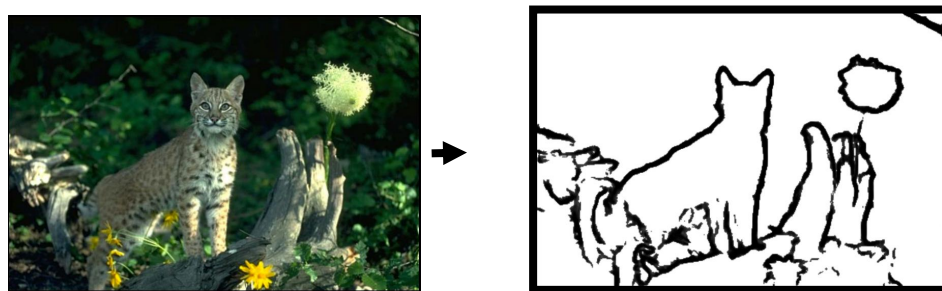


实例分割



[Long et al. 2015]

边缘检测



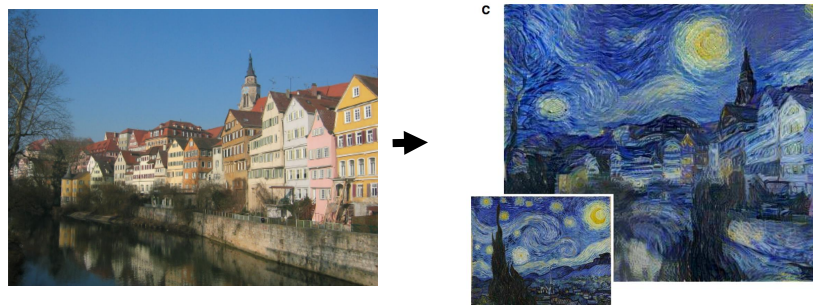
[Xie et al. 2015]

季节变化



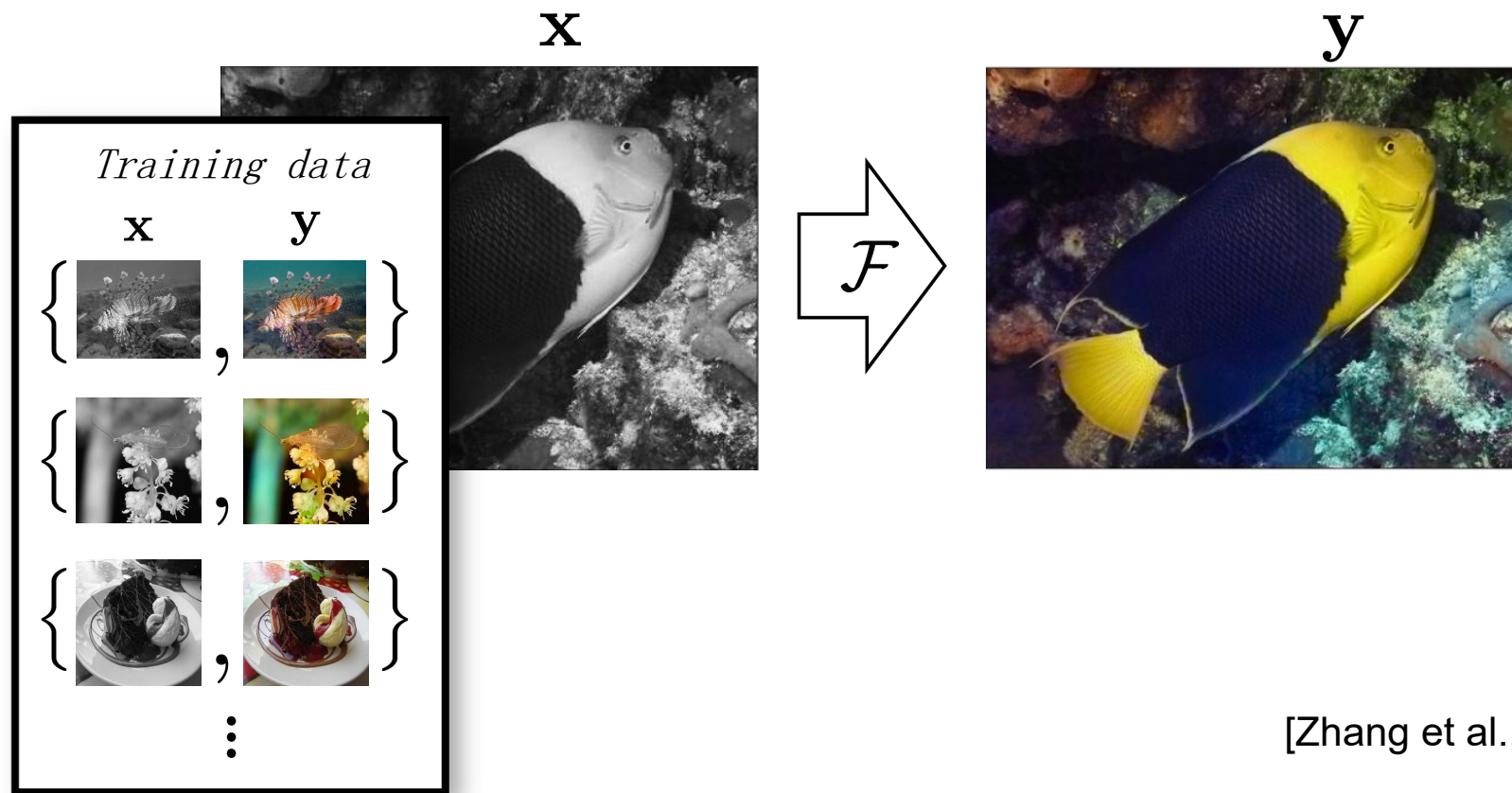
[Laffont et al. 2014]

艺术风格变化



[Gatys et al. 2016]

成对图像转换

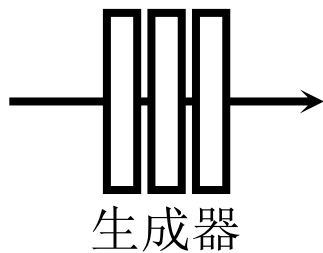


[Zhang et al., ECCV 2016]

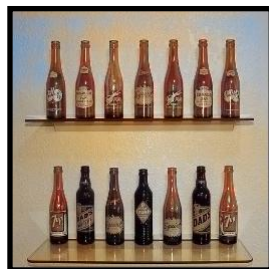
$\mathbf{x}$

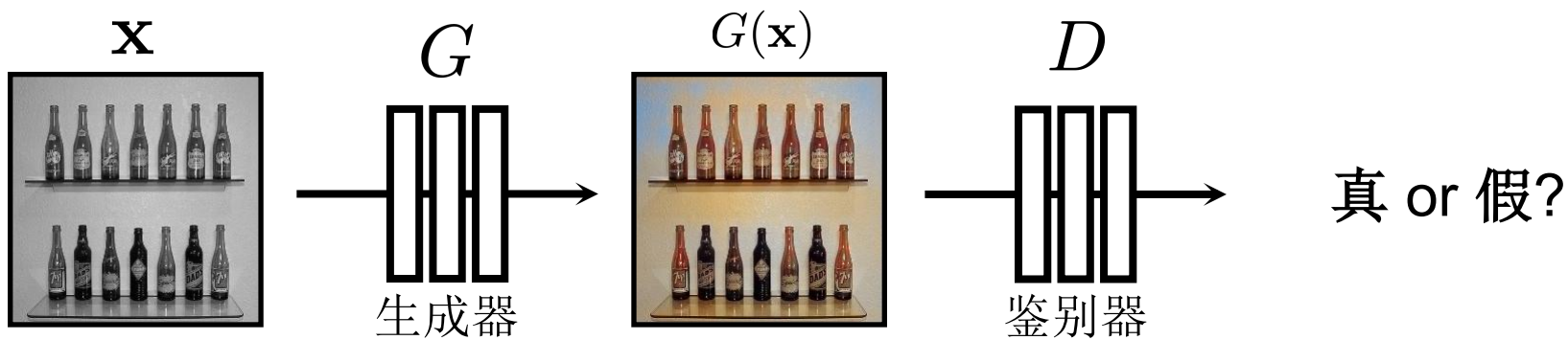


G



$G(\mathbf{x})$





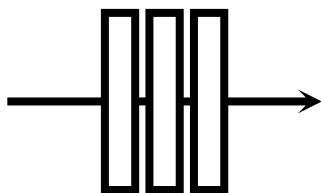
G 尝试生成假图片愚弄 D

D 尝试鉴别假图片

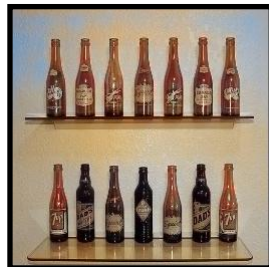
$\mathbf{x}$



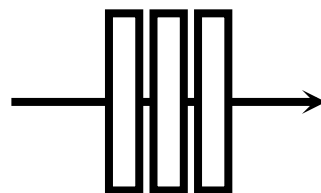
G



$G(\mathbf{x})$



D

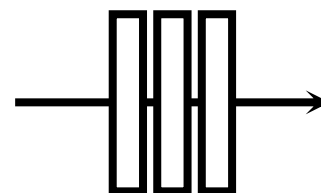


假 (0.9)

$\mathbf{y}$



D

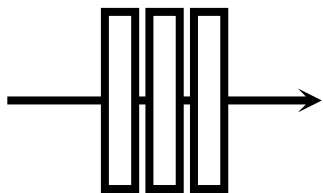


真 (0.1)

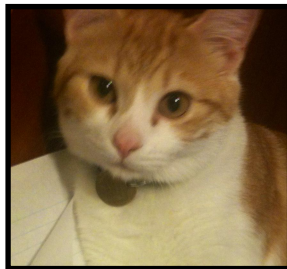
$\mathbf{x}$



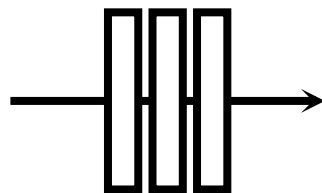
G



$G(\mathbf{x})$



D

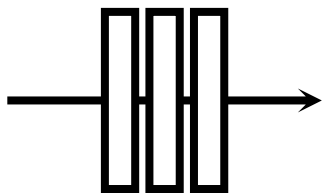


真!

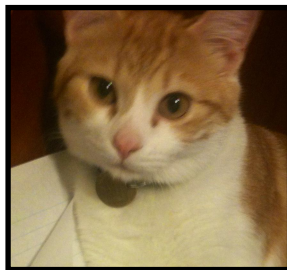
$\mathbf{x}$



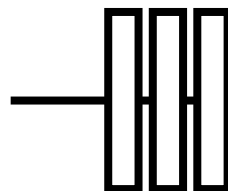
G



$G(\mathbf{x})$

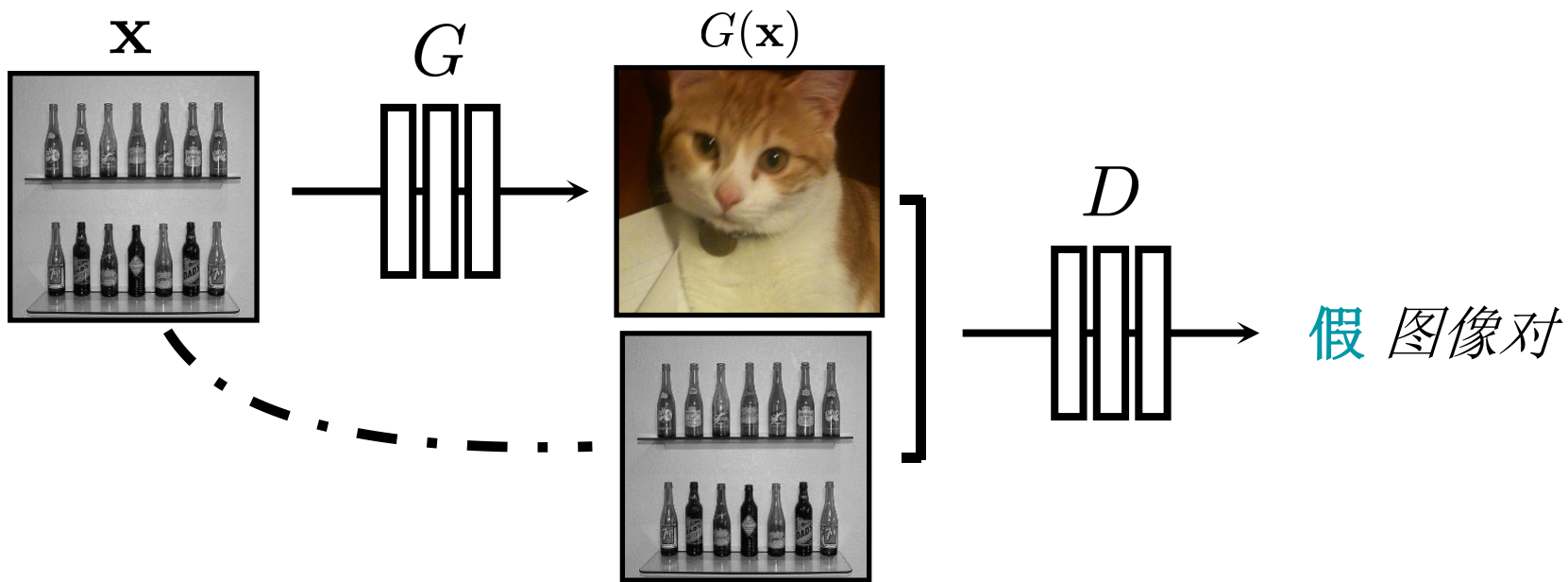


D

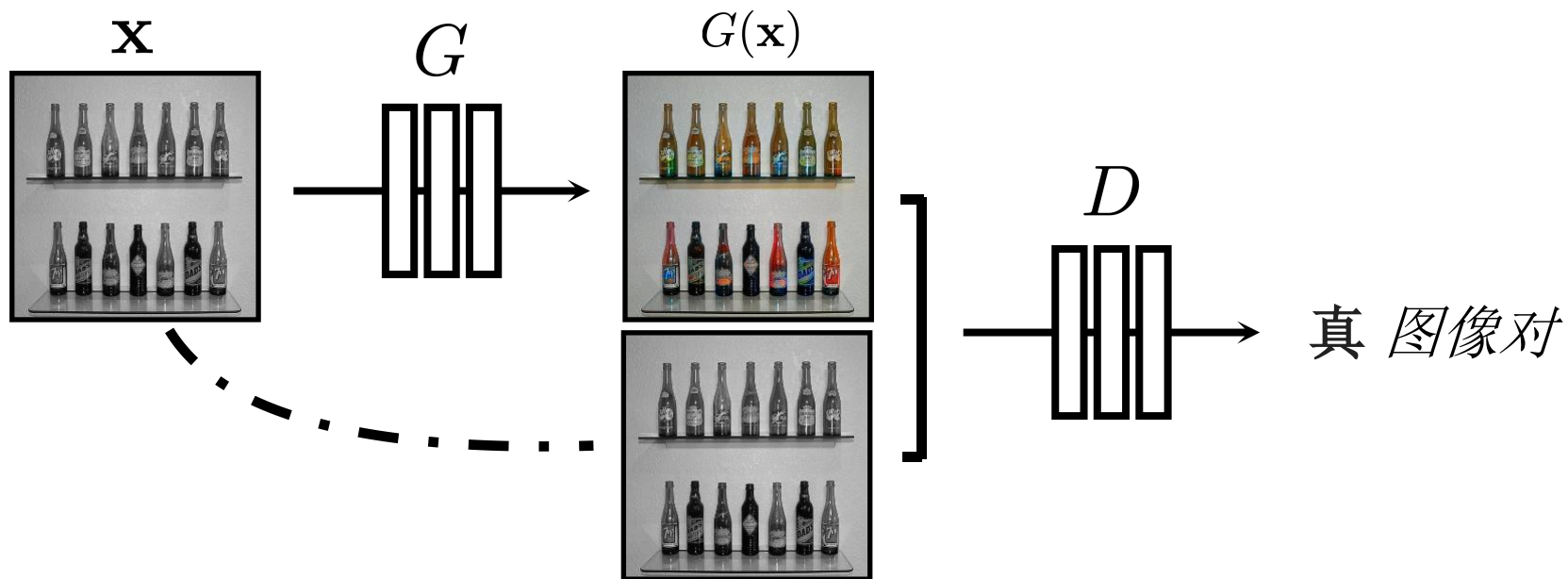


真 or 假 图像对 ?

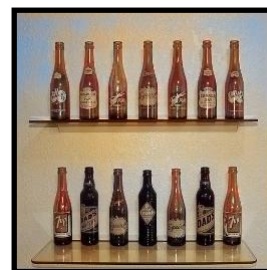




条件 GAN 损失



带条件损失



不带条件损失

灰度 → 彩色

输入

输出

输入

输出

输入

输出



数据来源[Russakovsky et al. 2015]

街道地图 → 卫星图片

输入

输出

真实分布

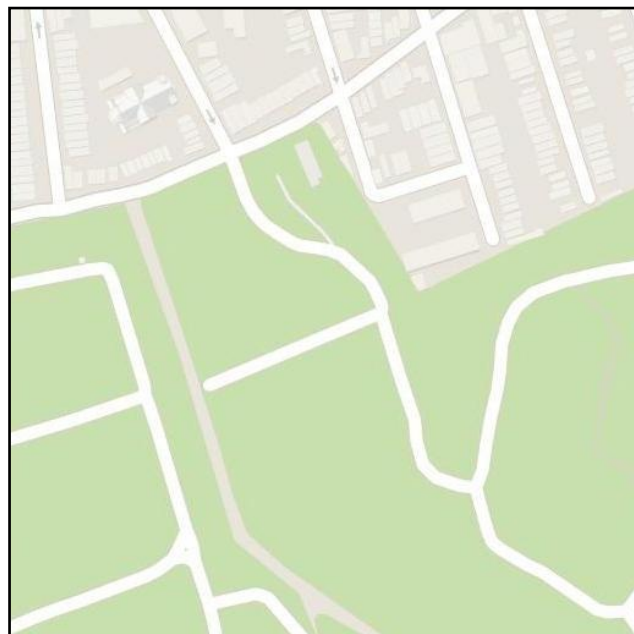


街道地图 → 卫星图片

输入

输出

真实分布

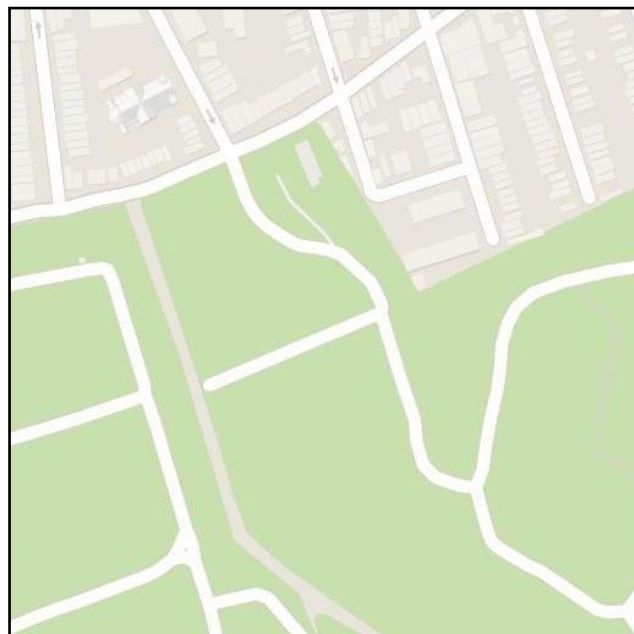


街道地图 → 卫星图片

输入

L1 输出

真实分布



成对图像对转换总结

- 转化图像 A 到 B
- A 和 B 有相似的结构
- 条件 GAN: 输出与输入是否关联?
- GANs 能够学习细节, L1/L2 不能.

GAN Vs 条件GAN

- GAN

$$\min_G \max_D V(D, G) = \mathbb{E}_{\mathbf{x} \sim p_{\text{data}}(\mathbf{x})} [\log D(\mathbf{x})] + \mathbb{E}_{\mathbf{z} \sim p_z(\mathbf{z})} [\log(1 - D(G(\mathbf{z})))] \quad (1)$$

- 条件GAN

$$\min_G \max_D V(D, G) = \mathbb{E}_{\mathbf{x} \sim p_{\text{data}}(\mathbf{x})} [\log D(\mathbf{x}|\mathbf{y})] + \mathbb{E}_{\mathbf{z} \sim p_z(\mathbf{z})} [\log(1 - D(G(\mathbf{z}|\mathbf{y})))] \quad (2)$$

Fig 1 illustrates the structure of a simple conditional adversarial net.

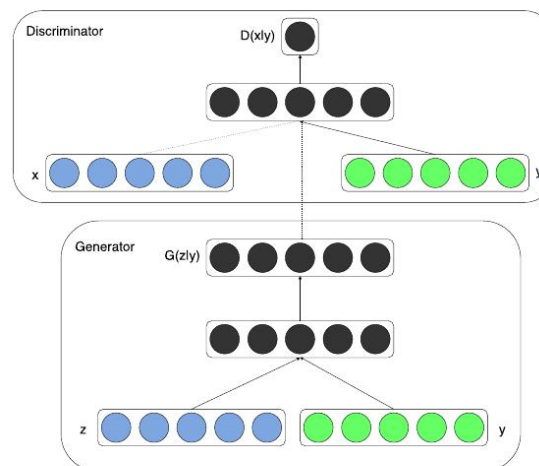


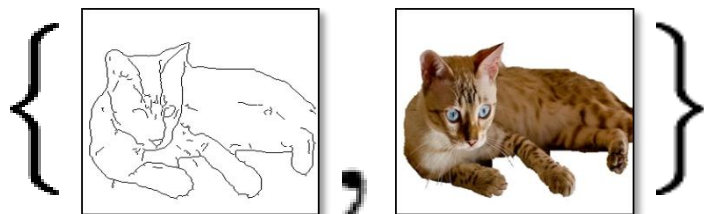
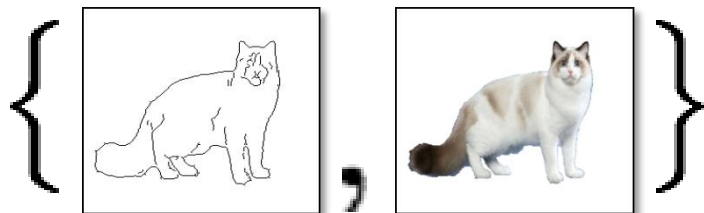
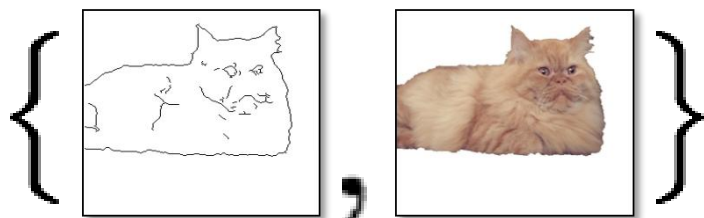
Figure 1: Conditional adversarial net

本章内容

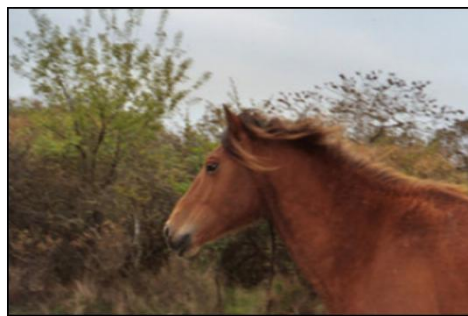
- 9.1.生成对抗网络定义
- 9.2.鉴别模型与生成模型
- 9.3.生成对抗式网络
- 9.4.条件GAN
- 9.5*. cycleGAN与styleGAN

不成对的图像对转换

成对

 x_i
 y_i


⋮



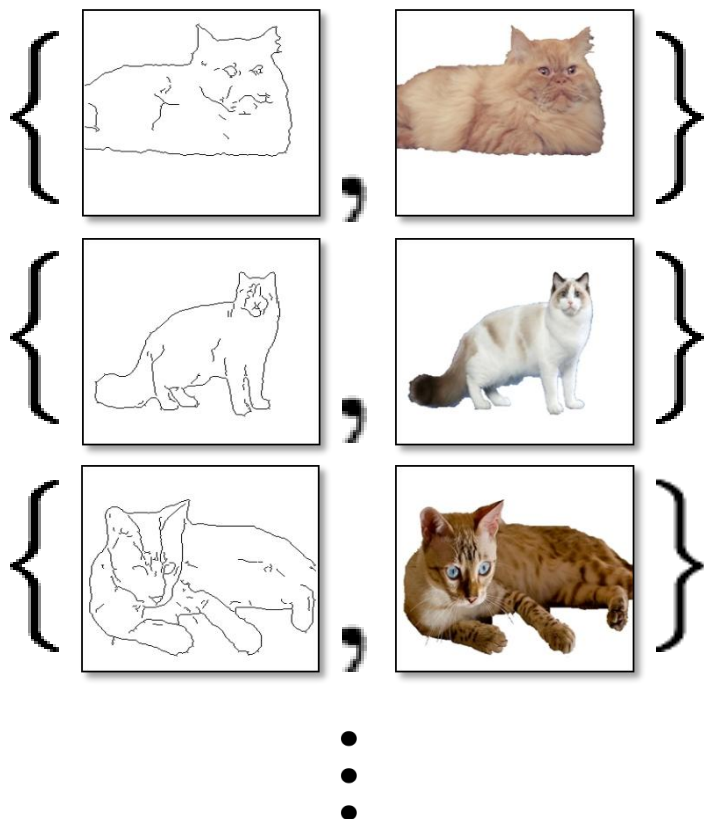
- 搜集图像对代价昂贵.
- 在很多场景无法使用.

[Zhu and Park]

成对

x_i

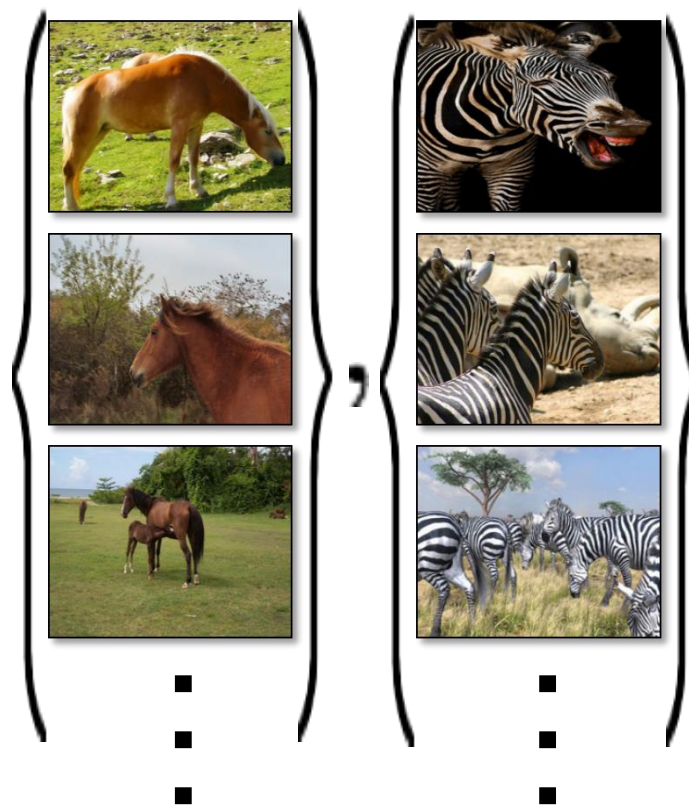
y_i

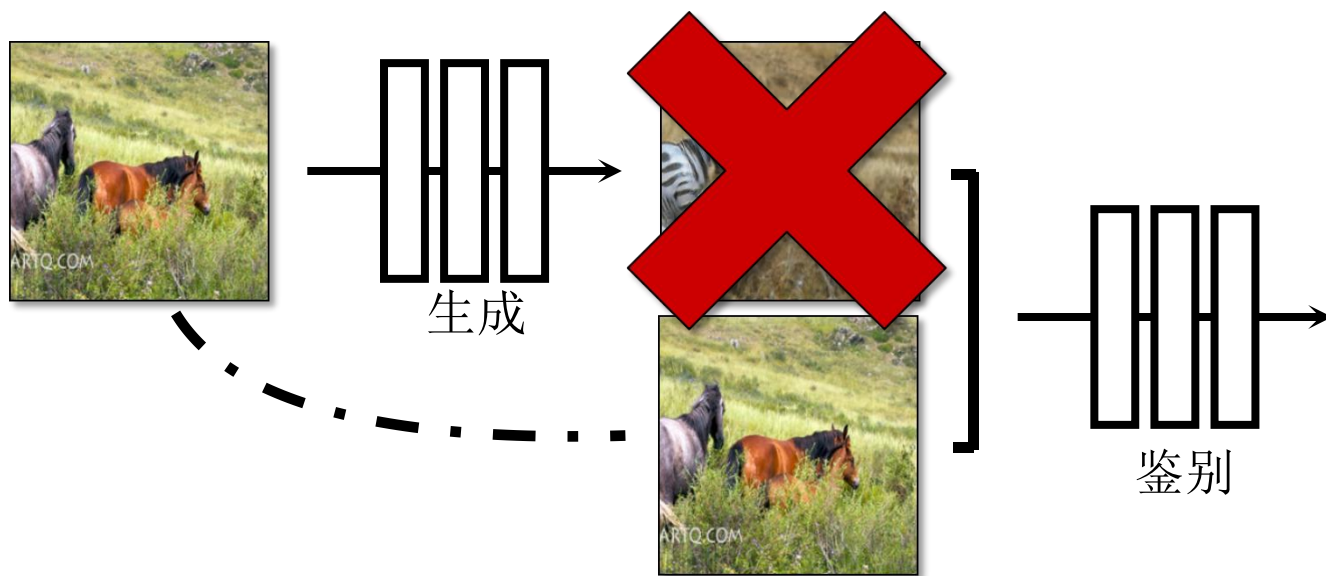


不成对

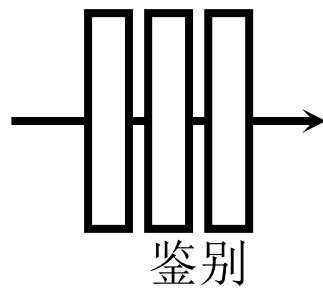
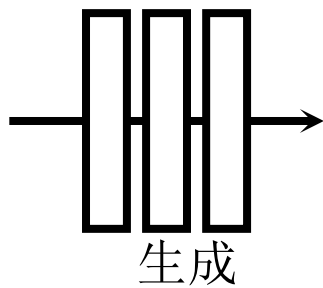
X

Y

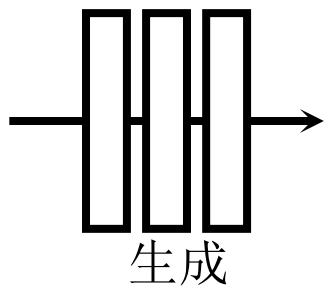




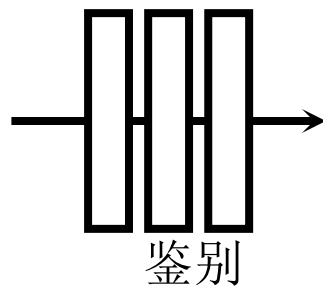
没有 输入-输出图像对!



真!



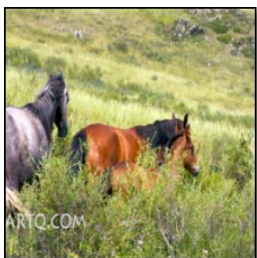
生成



鉴别

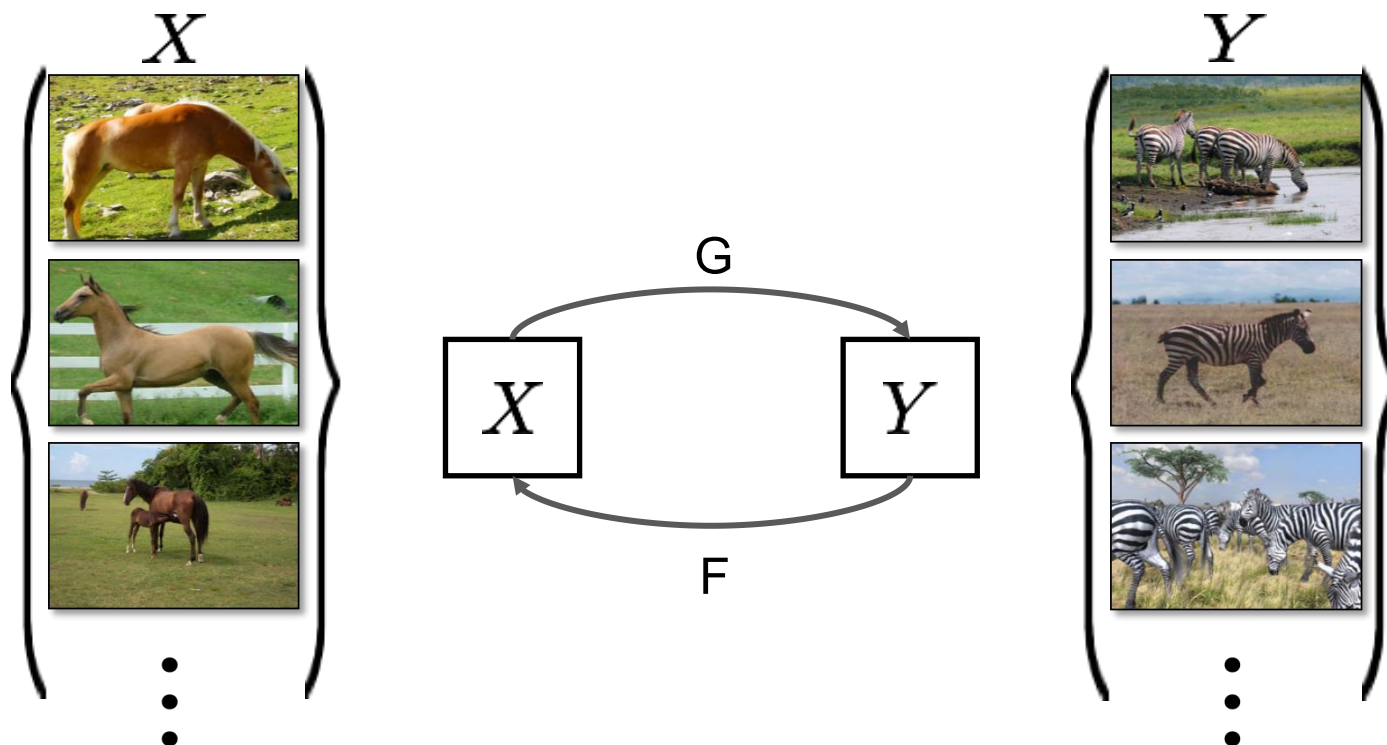
也是真!

GANs 不强迫输出与输入相关



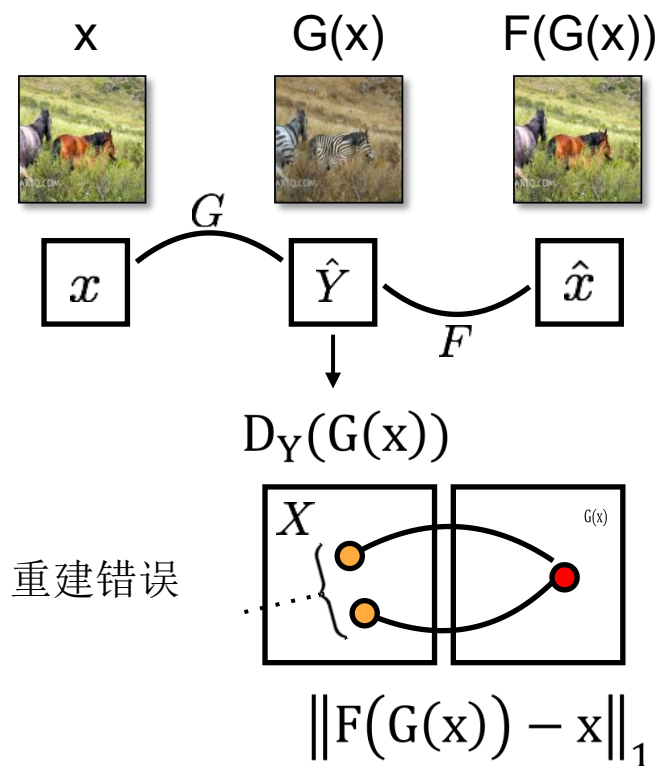
模型坍塌!

周期-一致对抗网络

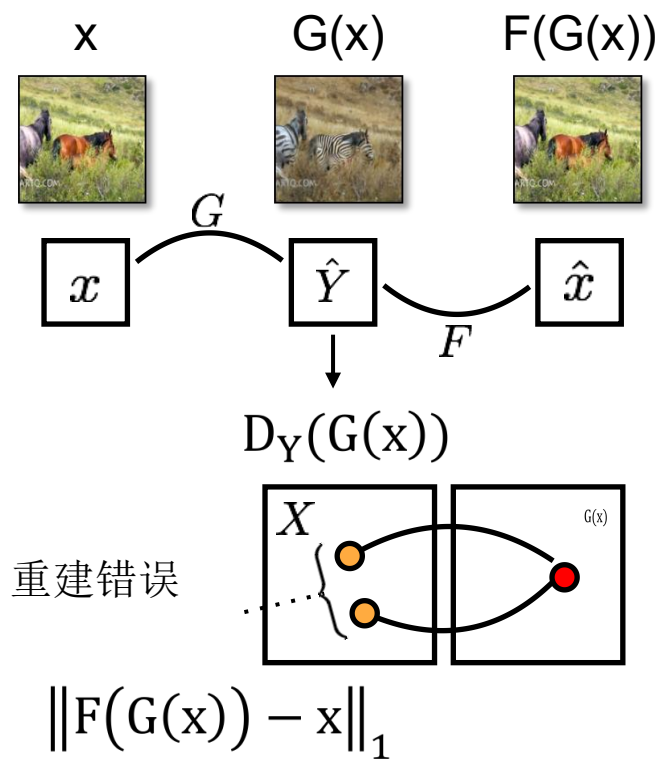


[CycleGAN](#): Jun-Yan Zhu, Taesung Park, Phillip Isola, Alexei A. Efros, ICCV 2017

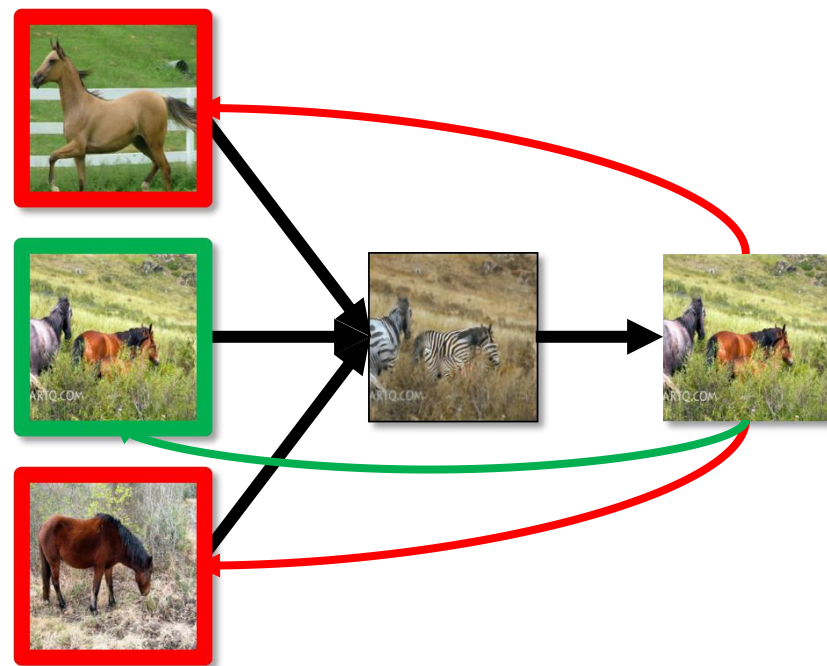
周期-一致对抗网络



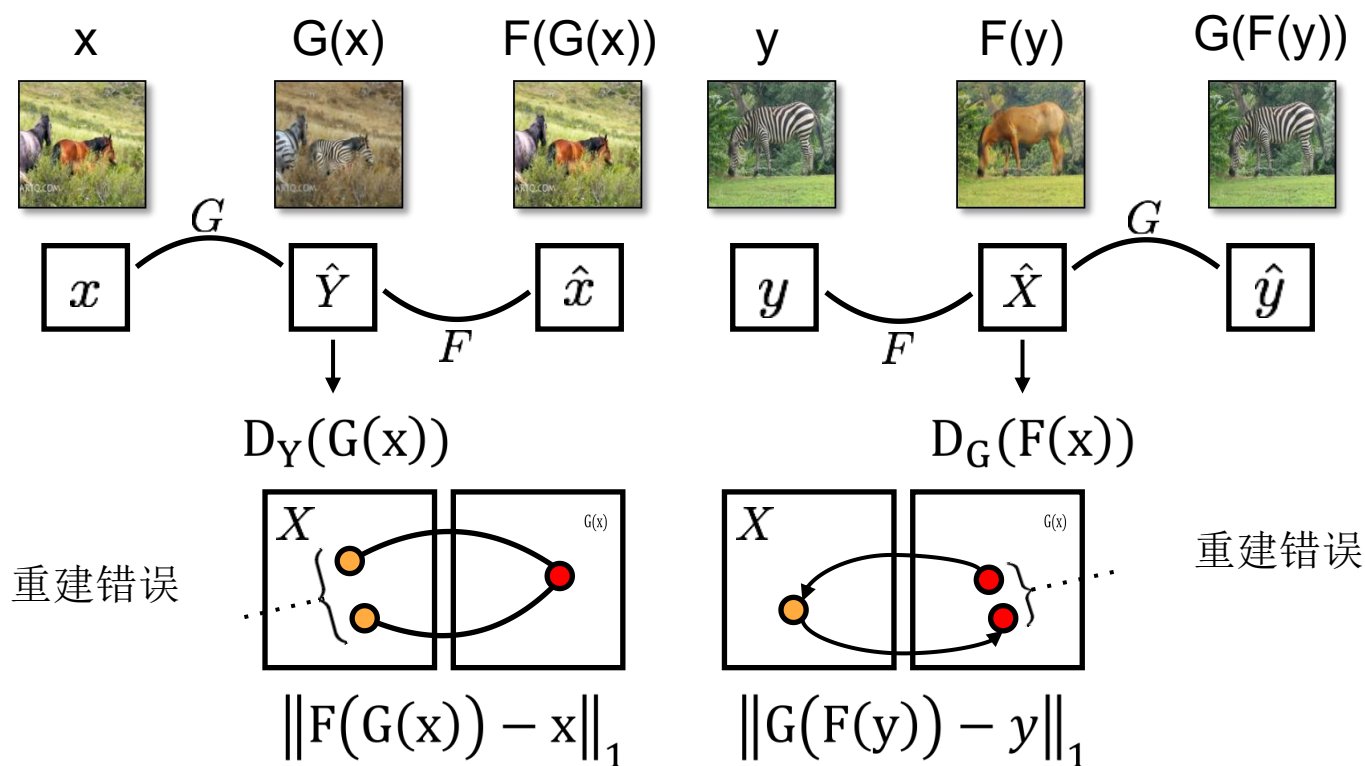
周期-一致损失



水 周期损失



周期-一致损失





[CycleGAN](#): Jun-Yan Zhu, Taesung Park, Phillip Isola, Alexei A. Efros, ICCV 2017⁸⁶



[CycleGAN](#): Jun-Yan Zhu, Taesung Park, Phillip Isola, Alexei A. Efros, ICCV 2017⁸⁷

印象派绘画到照片



[CycleGAN](#): Jun-Yan Zhu, Taesung Park, Phillip Isola, Alexei A. Efros, ICCV 2017⁸⁸

照片到印象派绘画

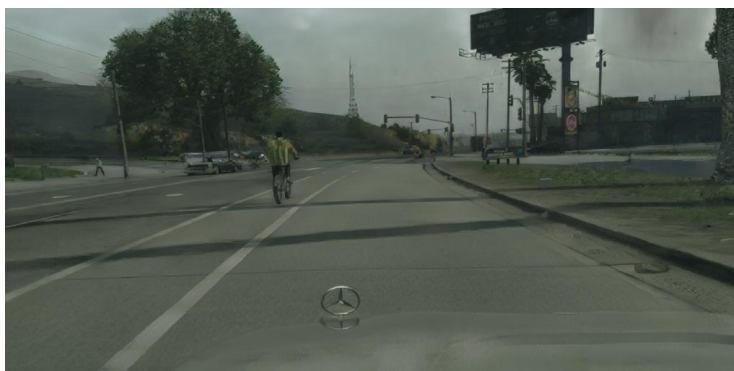


[CycleGAN](#): Jun-Yan Zhu, Taesung Park, Phillip Isola, Alexei A. Efros, ICCV 2017⁸⁹

使用GTA5训练自动驾驶汽车!



GTA5 CG 输入



输出



[CycleGAN](#): Jun-Yan Zhu, Taesung Park, Phillip Isola, Alexei A. Efros, ICCV 2017



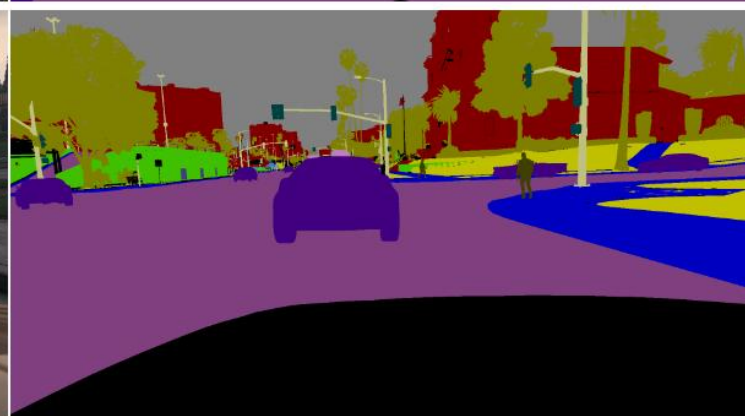
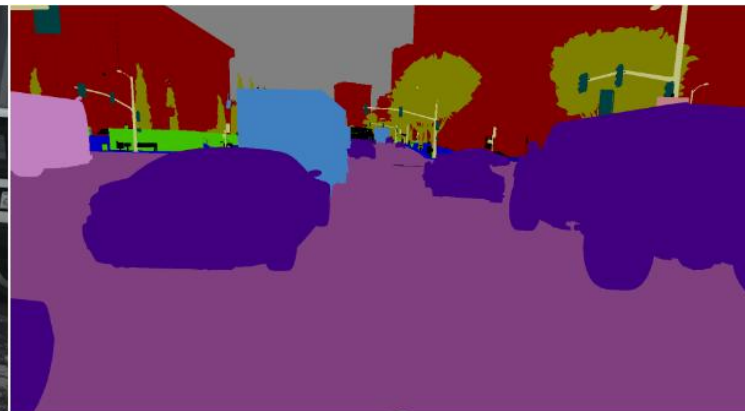
相片输入

输出

[CycleGAN](#): Jun-Yan Zhu, Taesung Park, Phillip Isola, Alexei A. Efros, ICCV 2017

合成训练数据

92



GTA5 图像

分割标签

[Richter*, Vineet* et al. 2016] [Krähenbühl et al. 2018]

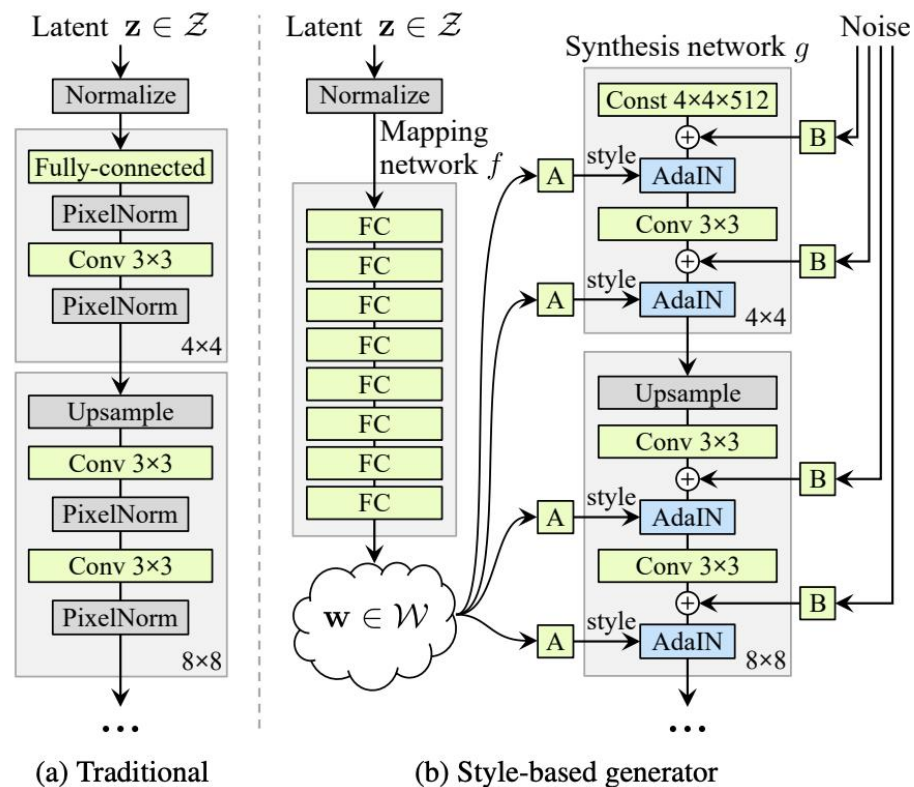
图像去模糊

93

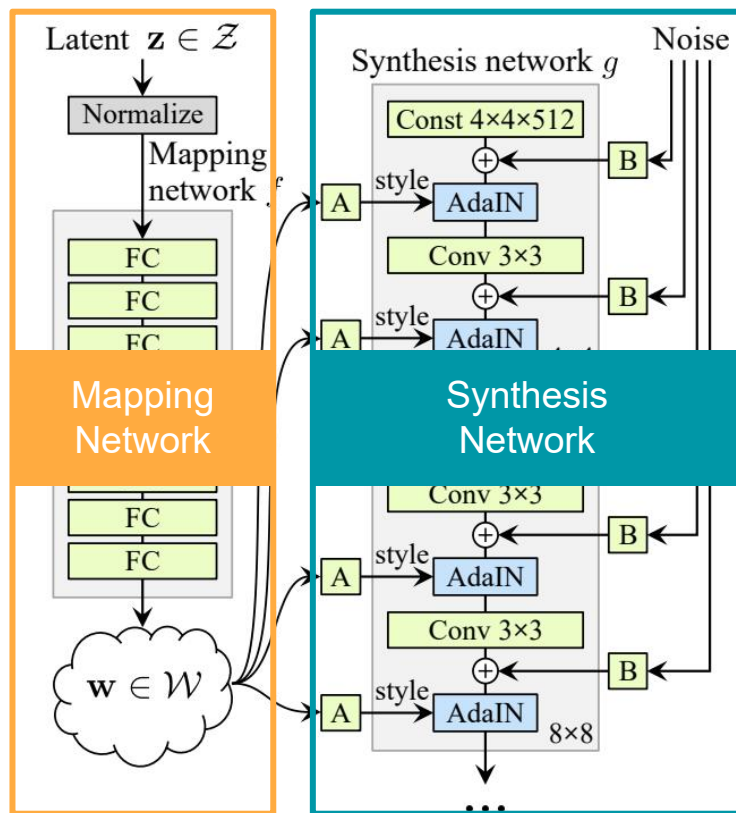


Cycle-Dehaze: Enhanced CycleGAN for Single Image Dehazing. CVPRW 2018
Deniz Engin* Anıl Genc* , Hazım Kemal Ekenel

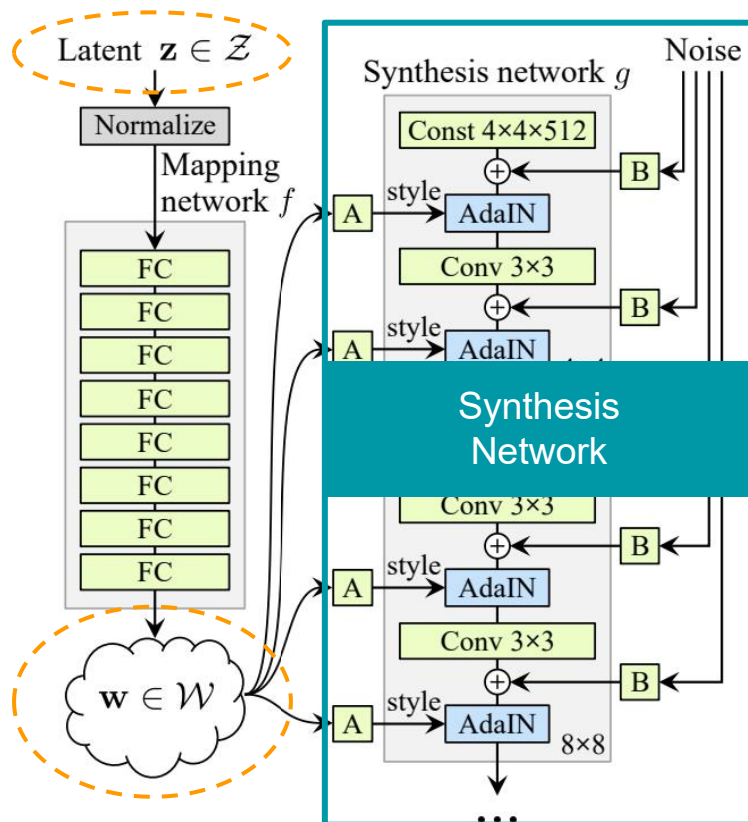
StyleGAN 结构



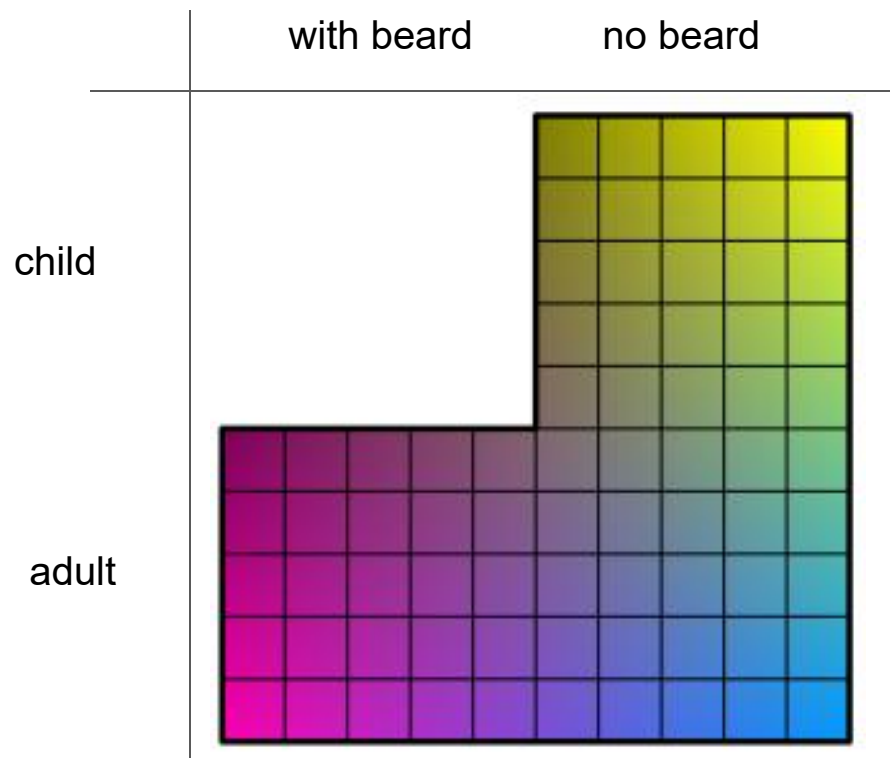
StyleGAN结构



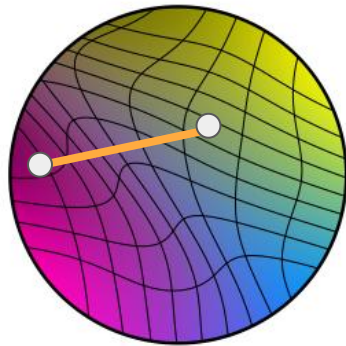
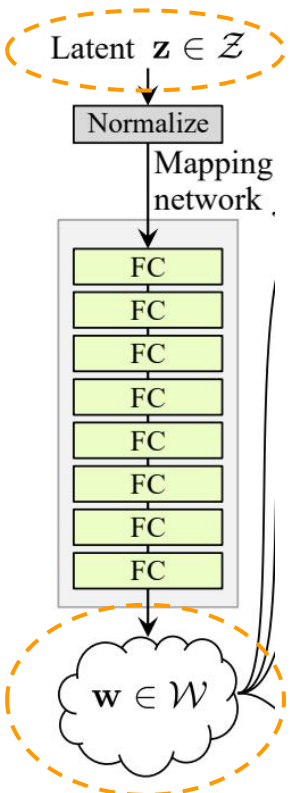
StyleGAN结构



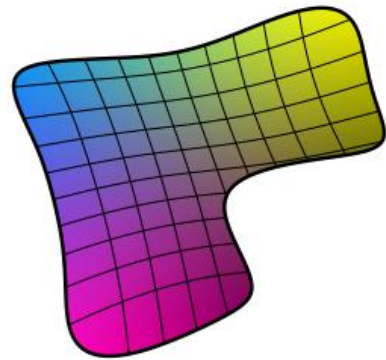
训练集中特征分布



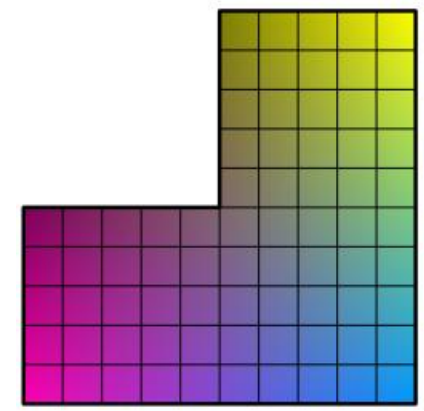
[Karras et. al., A Style-Based Generator Architecture for Generative Adversarial Networks]



(b) Mapping from $\mathcal{Z}$ to features



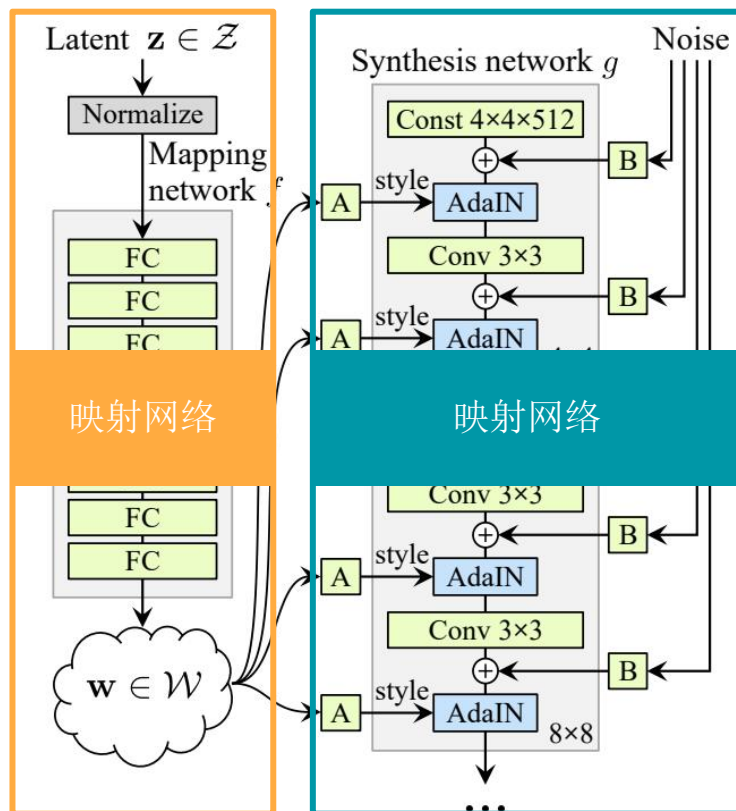
(c) Mapping from $\mathcal{W}$ to features



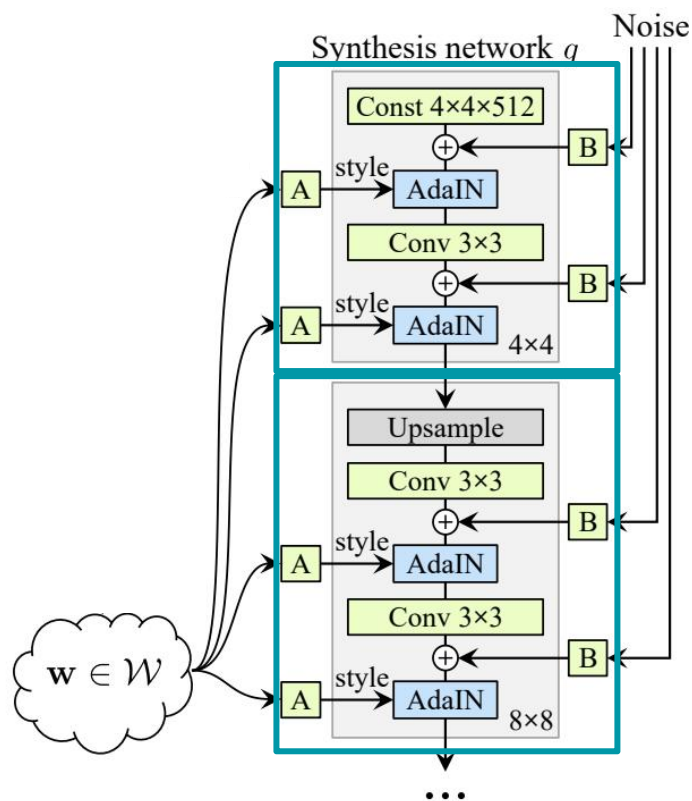
(a) Distribution of features in training set

[Karras et. al., A Style-Based Generator Architecture for Generative Adversarial Networks]

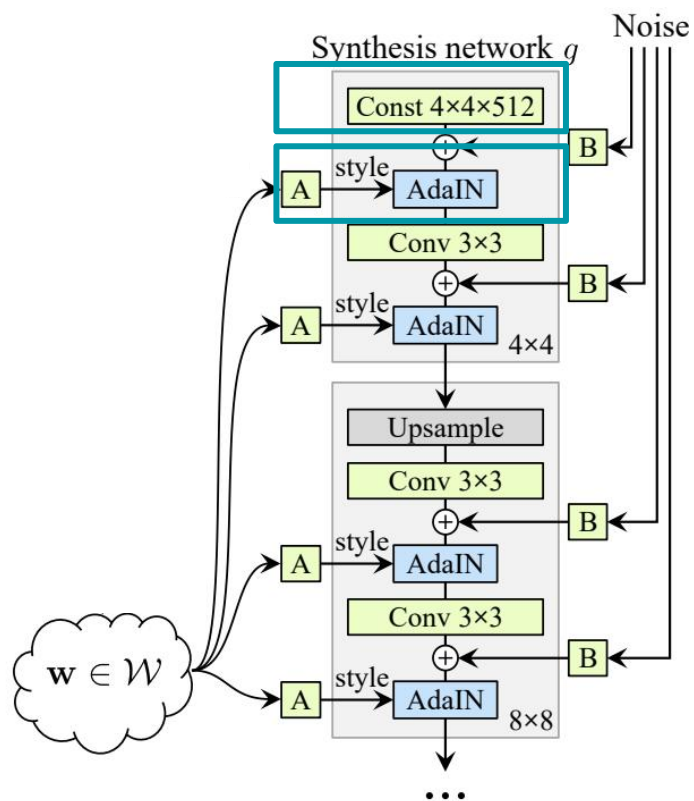
StyleGAN结构



StyleGAN结构



StyleGAN结构



[Karras et. al., A Style-Based Generator Architecture for Generative Adversarial Networks]

风格迁移





内容损失
[结构]

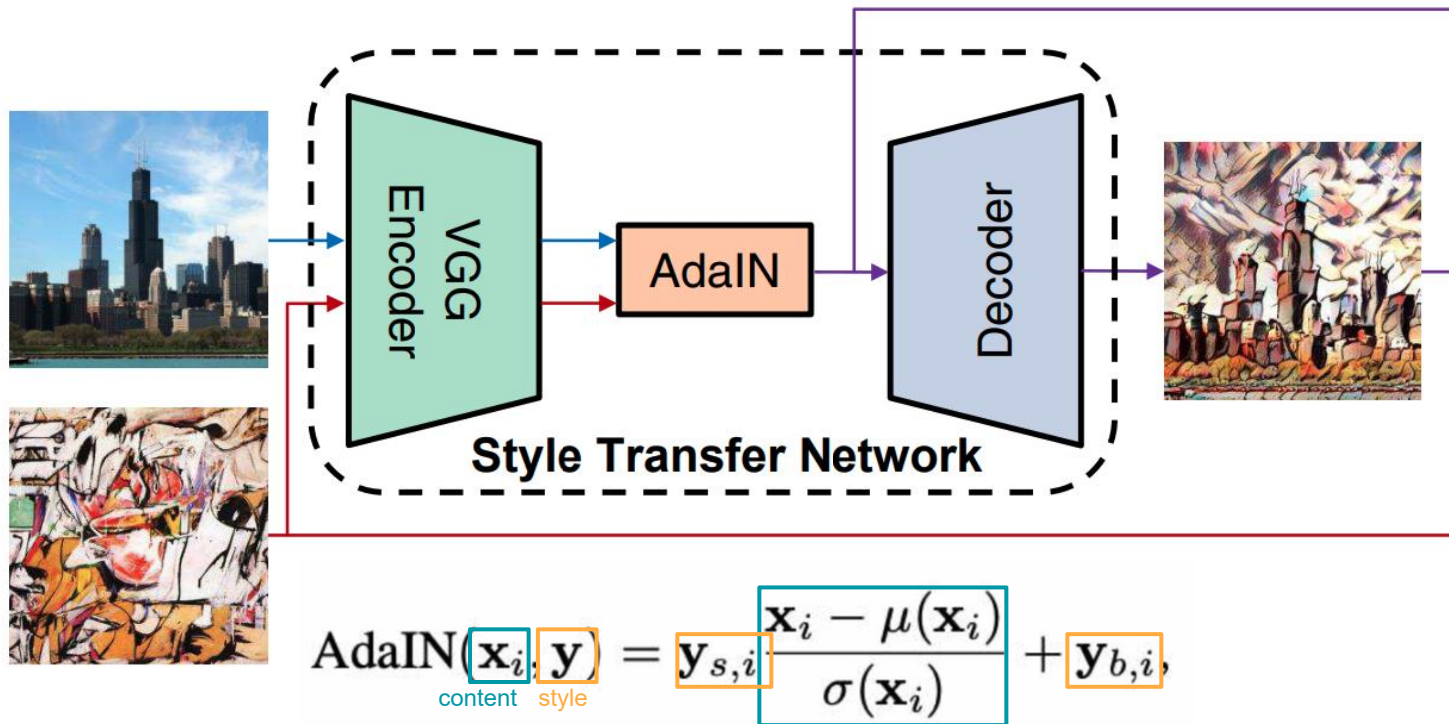


风格损失
[颜色, 笔画, etc.]

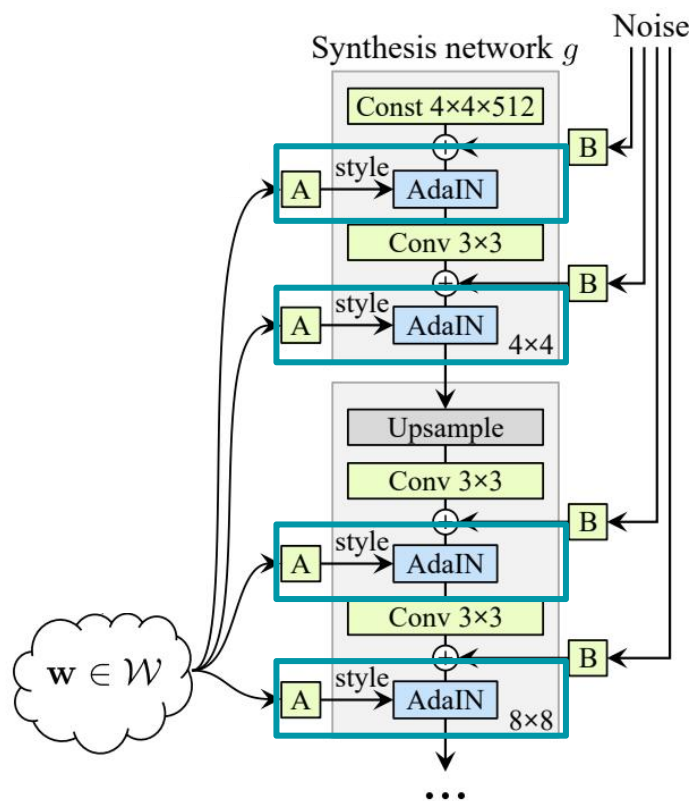


太慢!!!

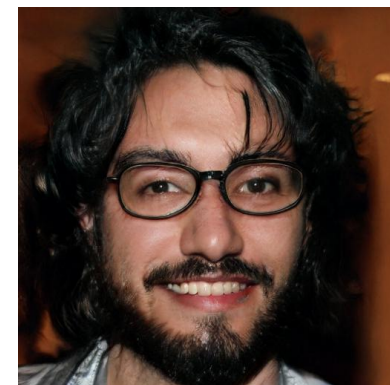
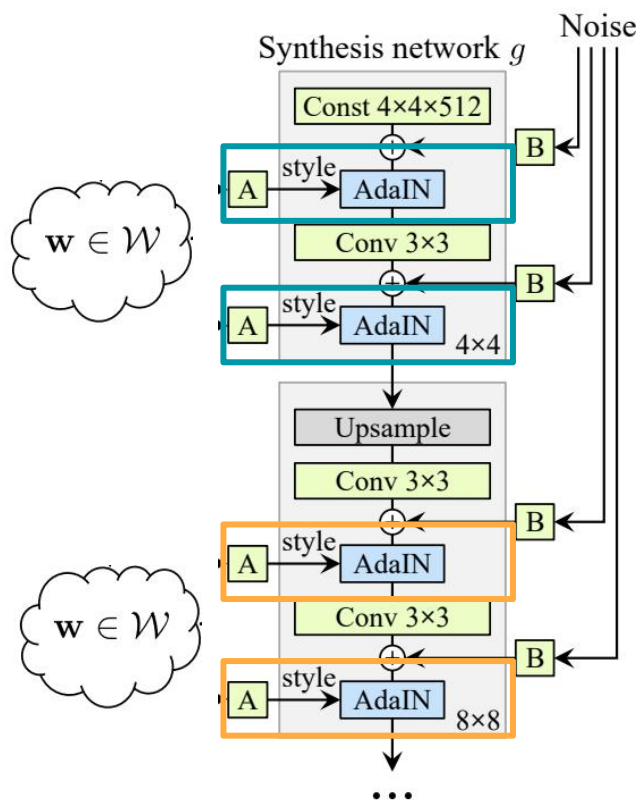
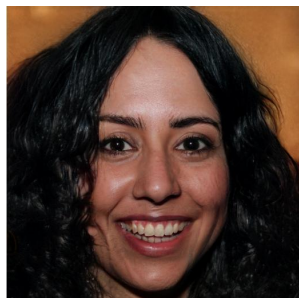


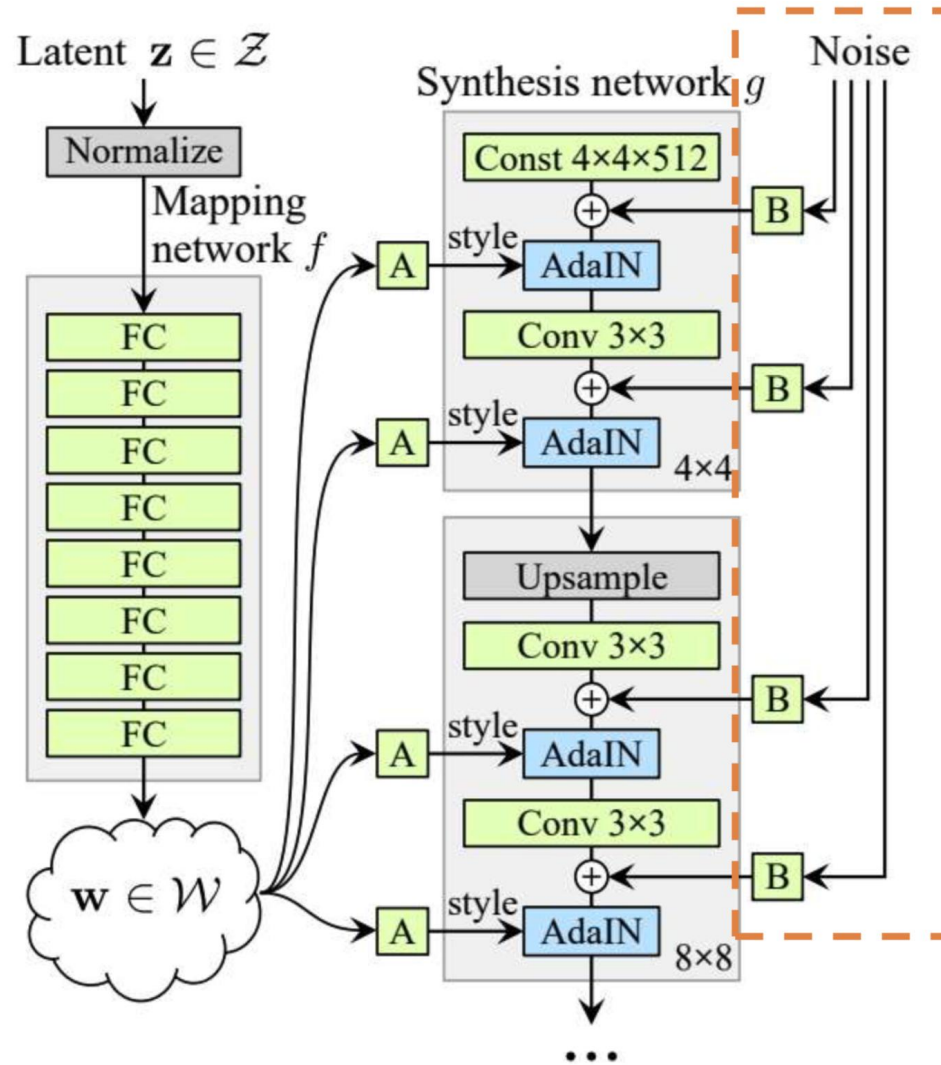


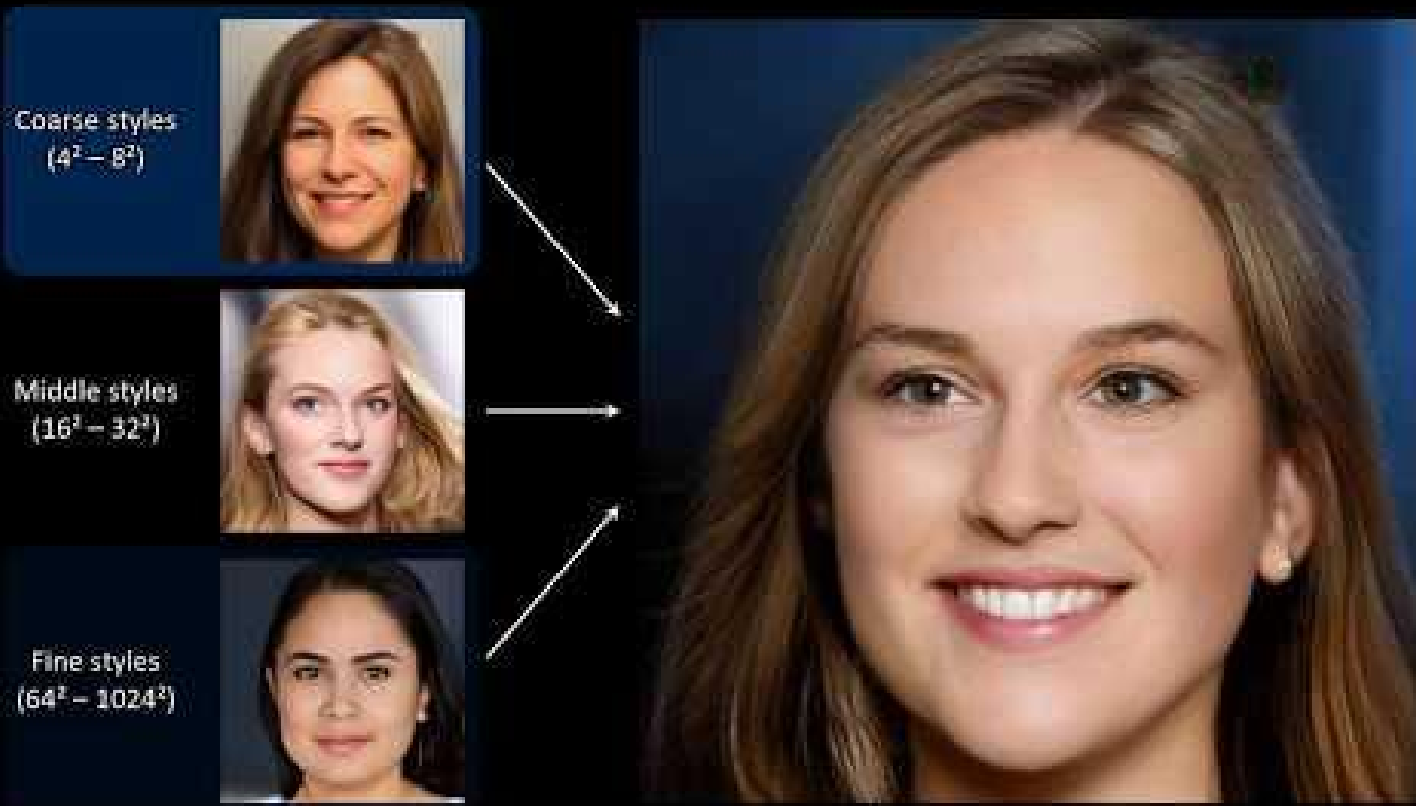
StyleGAN结构



StyleGAN结构







本章小结

- 生成对抗网络定义
- 鉴别模型与生成模型
- 生成对抗式网络
- 条件GAN
- cycleGAN与styleGAN

结束

下一次课

- 图像对齐与拼接